

THE NEW LANDSCAPE

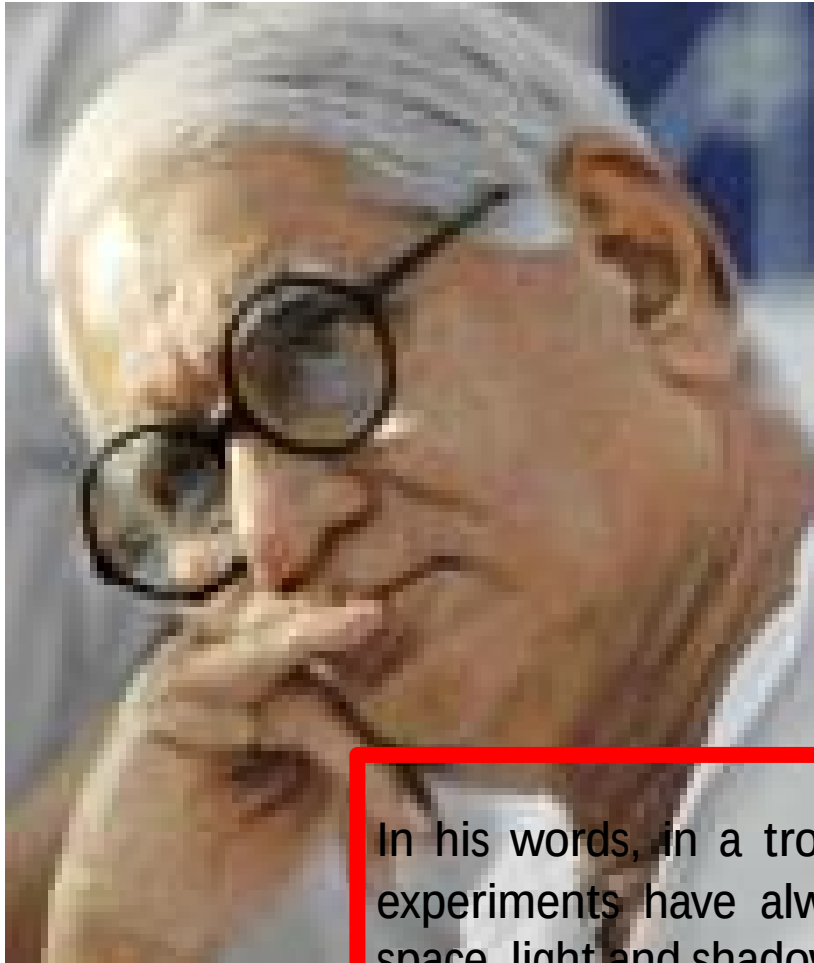
URBANISATION IN THE THIRD WORLD

Charles Correa



A Mirar Book

Butterworth Architecture



In his words, in a tropical country, "space is a resource" .His experiments have always made full use of the ambiguous nature of space, light and shadow, in a tropical climate.

"Improving the habitat needs visual skills" writes Charles Correa in his blueprint for change, *The New Landscape* in which he has analyzed the problems facing the urban landscape and provided his suggestions.

The lack of planners and the role of architects as experts, rather than executioners in policy and decision making also undermines the proper maintaining and development of a city, he mentioned. Adding further that skyscrapers were not a solution to end the problem of housing in cities like Mumbai and Pune, architect Correa endorsed the traditional way of building houses as they were affordable and provided better living conditions.

In many of his published articles he has explained how low rise typical developments in countries like India, Pakistan or Bangladesh can achieve a certain density in urban development. By his word high density area doesn't mean containing high-rises. Rather high-rises often reduce the job opportunity of unskilled floating urban population.

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01 urbanization



Planning for Bombay.
1964.



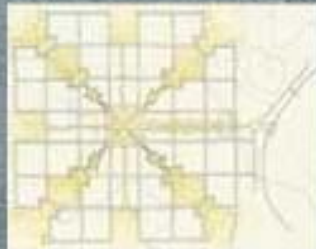
Hawkers Pavements.
Bombay. 1968.



Cochin Waterfront.
Cochin, Kerala. 1974.



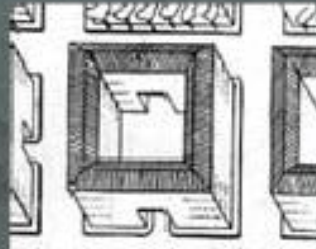
The National Commission on
Urbanisation. India. 1985 - 87.



New Bagalkot.
Karnataka. 1985 - to date.



Ulwe, New Bombay City
Centre. 1991 - till date.



Ulwe: Housing.
New Bombay. 1991 - till date.



Ulwe: Urban Design.
New Bombay. 1991 - till date.



Redevelopment Parel Mills.
Bombay. 1996 - till date



Buddhapurnima Lakefront.
Hyderabad. 1999 - till date.



Salt Lake City Centre.
Calcutta. 1999 - 2004.



New phenomena



Mass migration in urban areas

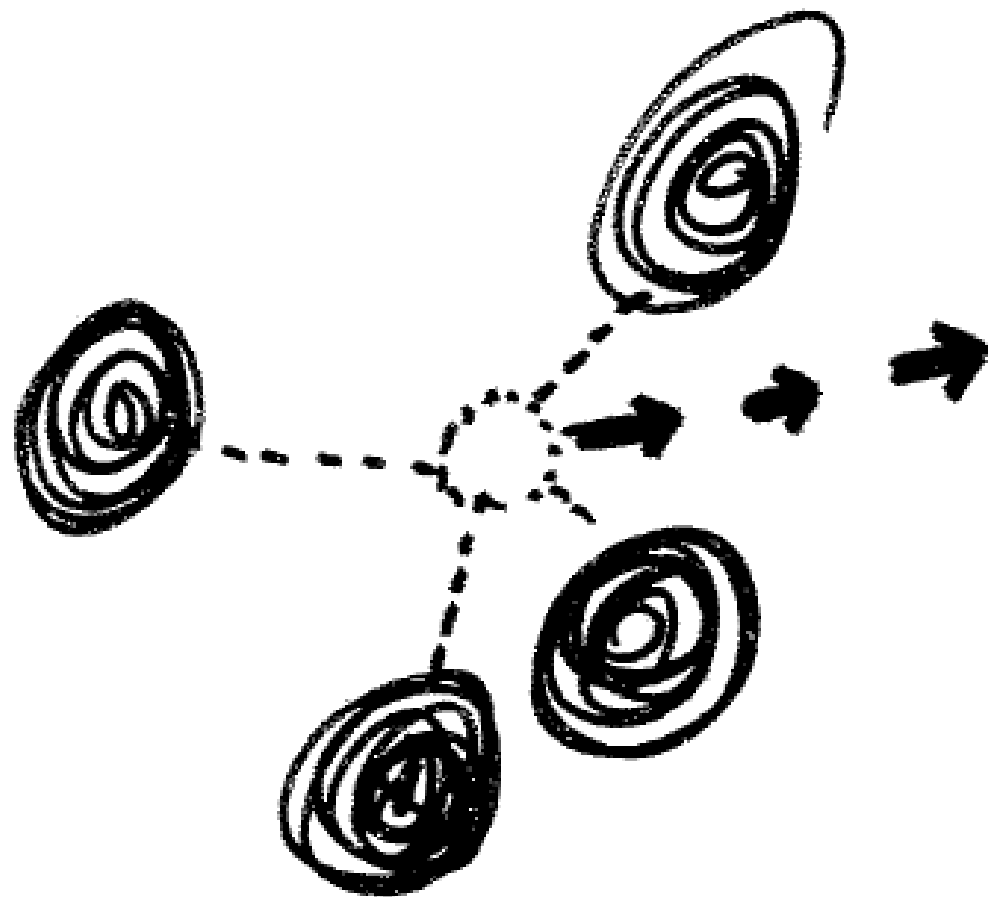
Squatter : anti social element

Urban center :growing twice as fast as over all population

Distress migration: pull of big city lights-marginal earners [land less labor]

Housing: a very low priority on their list of needs.

they want to be where their job are.



Village cluster growth: New employment (cottage industries, etc.) and social infrastructure (a health centre, a school) located centrally to a cluster of villages. Inhabitants continue to use existing stock of housing until the whole cluster gradually coalesces into a new growth centre. This would also allow the rationalisation of the transport system: cycle and bullock-cart from village to new centre, buses between centres, etc. all the way up to trains and aeroplanes connecting the large metropolii.

Solutions are complex



Distress migration

Solution 01: land redistribution and social reform in villages



Increase holding capacity

Solution 02: identify key market towns, appropriate investment



New growth centers

Solution 03: all new industry ,Government buildings



Locate on small and middle sized cities

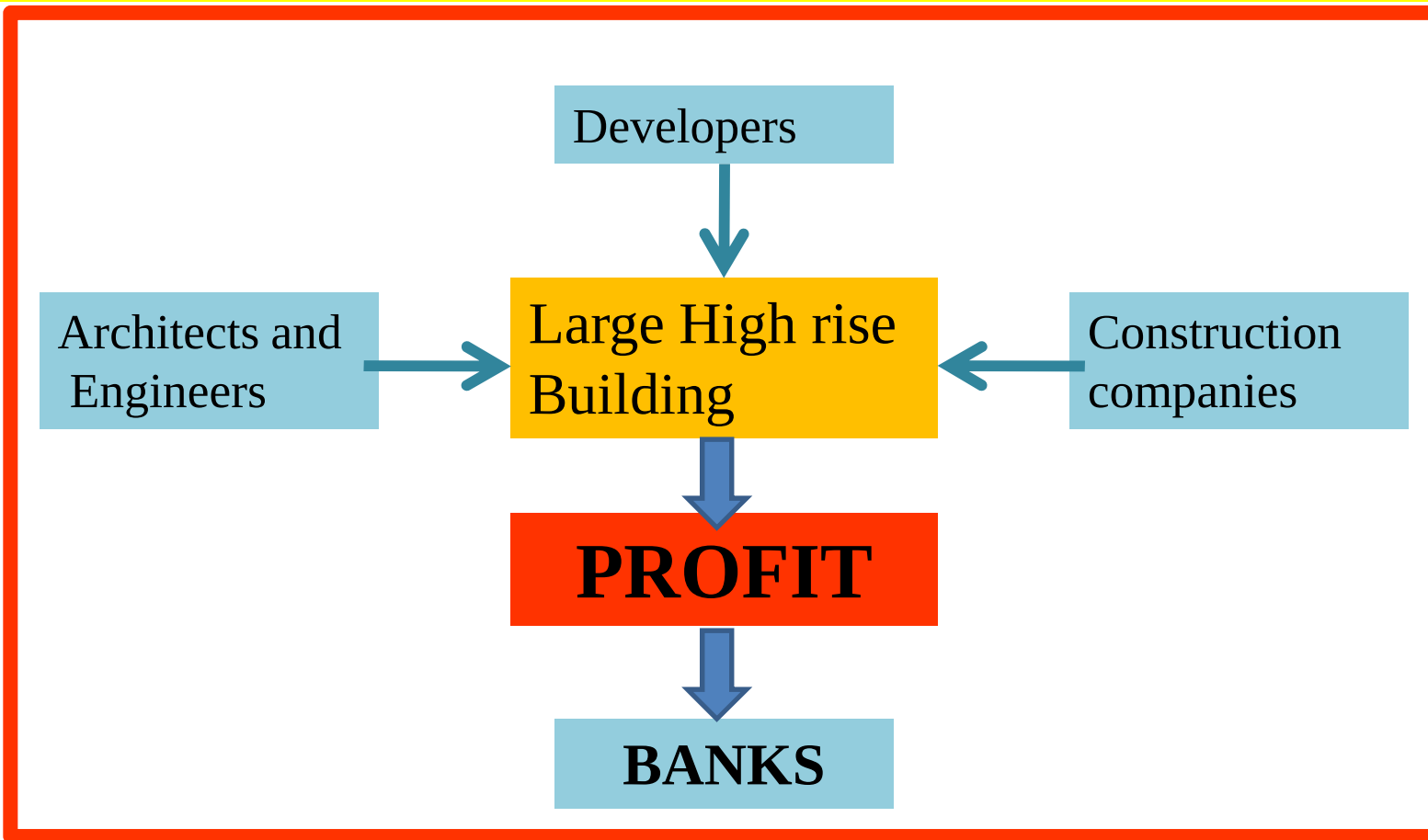
Restrictive monopolies: high rise building.



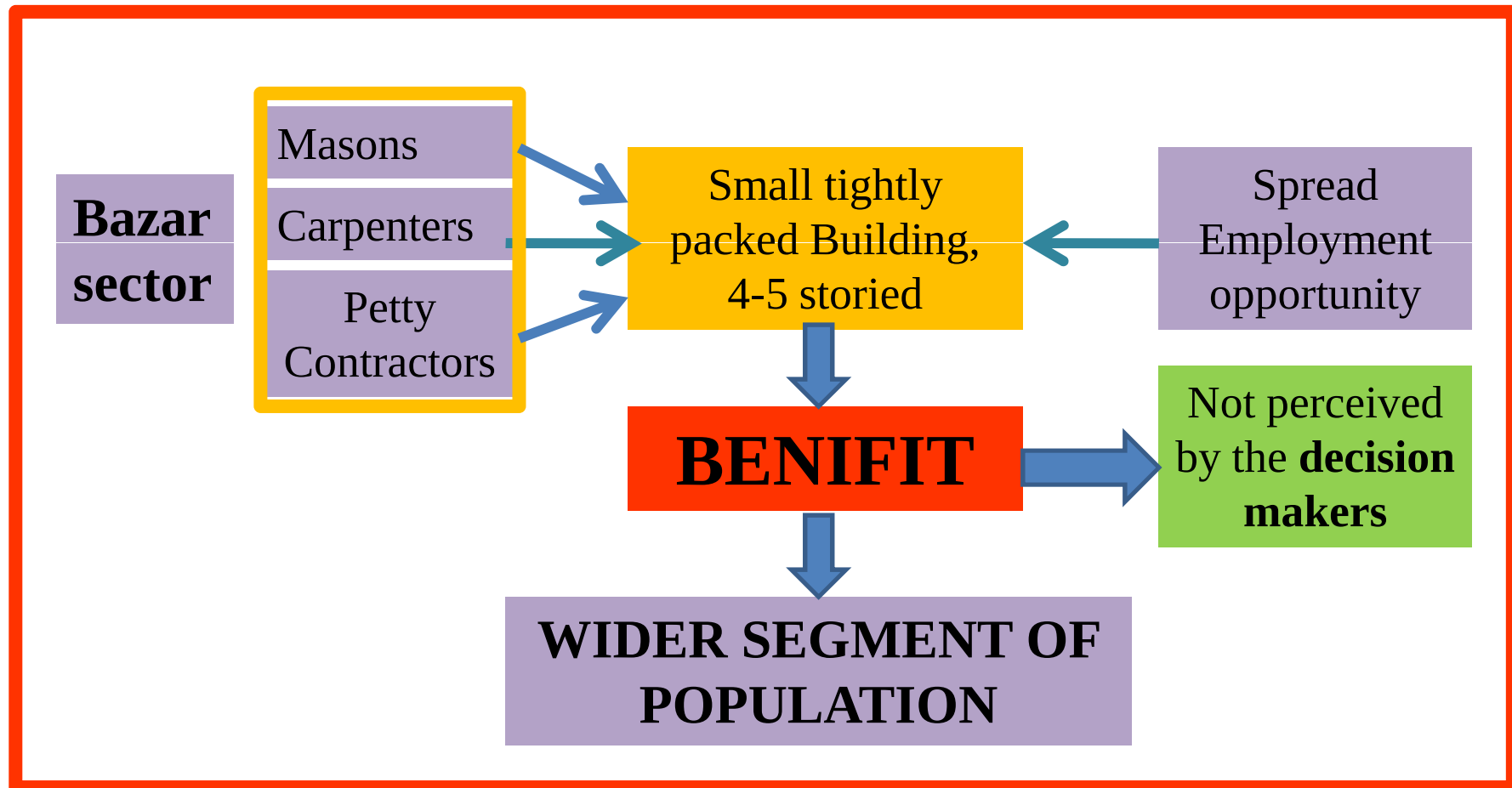
Low-rise, high-density built form: generating jobs in the bazaar.



Crucial role of our town and cities



Pattern of development in old city centers of third world

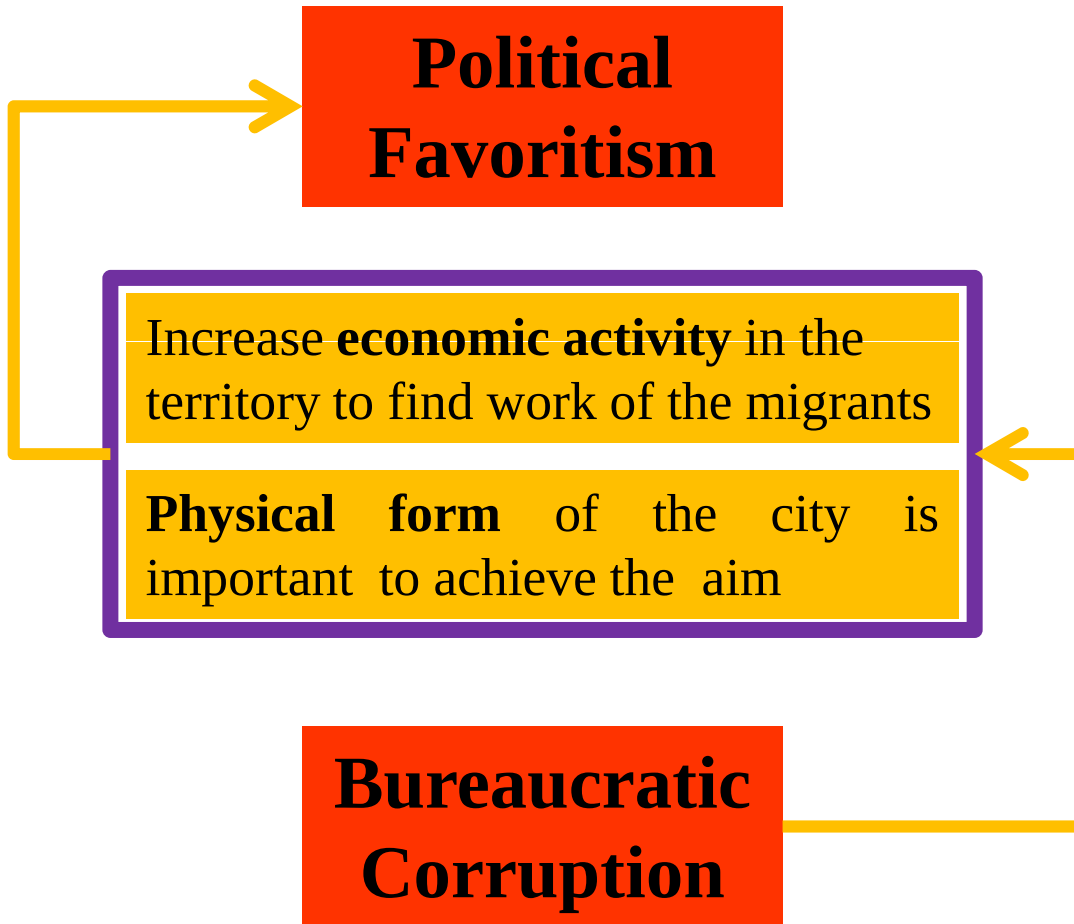


**Political
Favoritism**

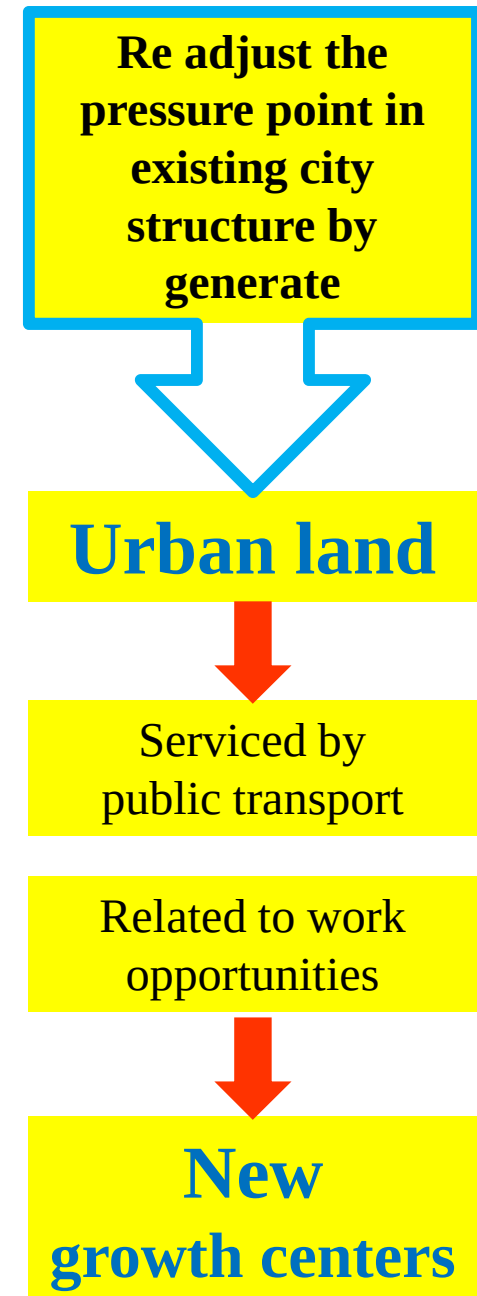
Increase **economic activity** in the
territory to find work of the migrants

Physical form of the city is
important to achieve the aim

**Bureaucratic
Corruption**



problem, there does not seem to be any solution in sight. To find the new landscape, we must start with an overview; we must examine the *entire* system we call “city” and try to identify those living patterns, those life-styles, which are optimal in their totality — including roads, services, schools, transportation systems, social facilities and, of course, the housing units themselves. Only then will we be able to perceive how one can, in Buckminster Fuller’s ineffable phrase “rearrange the scenery”



Most Third World primate cities (Calcutta, Lima, Hong Kong, etc) came into being as an interface between colonial powers and the hinterland. The colonials developed them for their own ends – on a scale and with an economic and physical structure that suited their purpose.

After independence, these cities entered a period of rapid growth with no upper limit in sight. In most cases, the new national governments have paid little or no attention to adjusting the obsolete city-structures. Hence, their holding capacity has no relation to the scale of the demands being made on them.

There is indeed a brutal mismatch between the city's structure and the load it must carry today. As the population escalates, competition for the same facilities increases. And of course the rich win out, pre-empting desirable urban land. Compare the pattern of job locations in the city with land prices. Is it any wonder that the poor sleep out on the pavements?

Re –structuring the CITY

02

space as resource

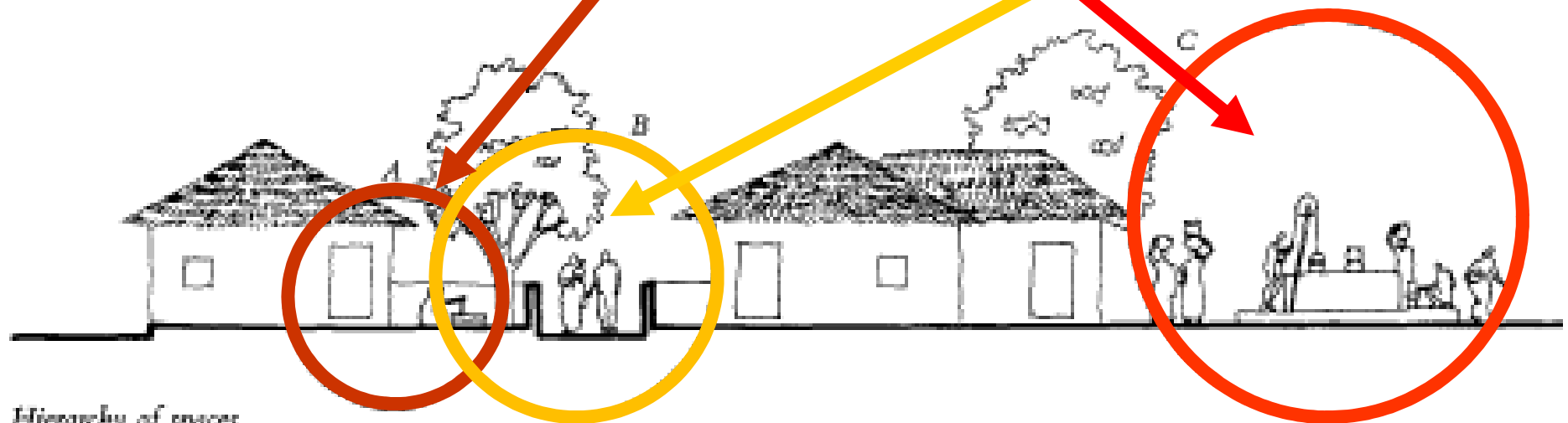


Maidans are pre-empted by sportsmen ...



Key to Correa's design approach is his understanding of the unique cultural requirements and needs of his native country. He recognizes that living in an Asian city requires much more than the use of a small room. In a 1987 book focused on his work, Correa said, "Such a cell is only one element in a whole system of spaces people need in order to live."

He identifies this hierarchical system as consisting of **four major elements**: space needed by the family for private use **[A]**, areas of intimate contact **[B]** (e.g. the front doorstep where children play), neighborhood spaces **[C]**, like a water tap, and urban area open space **[D]** used by the whole city.



Hierarchy of spaces.

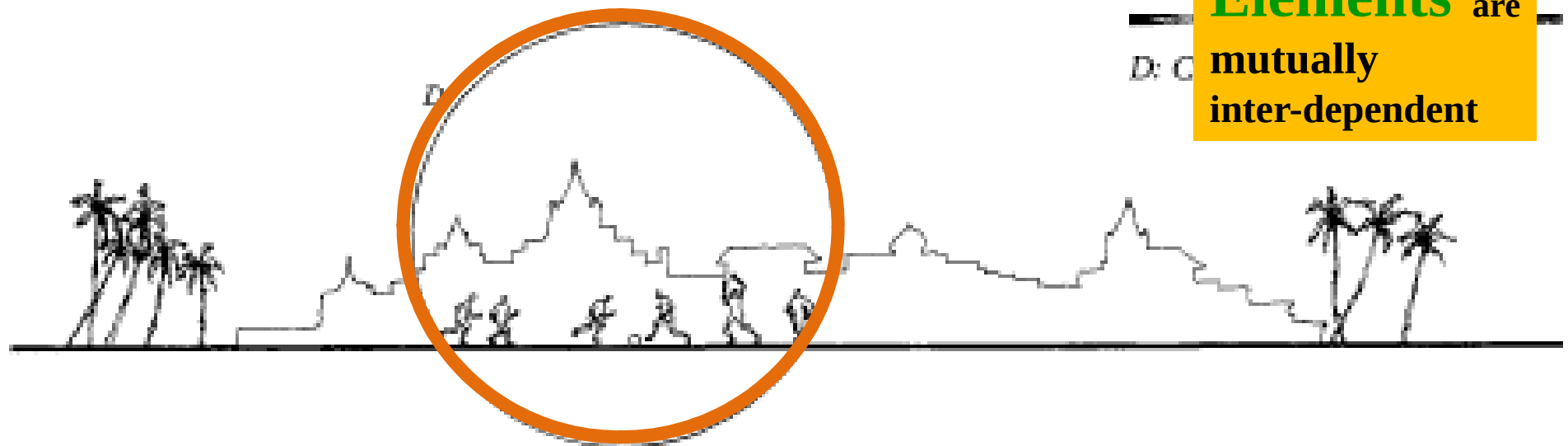
In different societies the number of elements and their inter-relations may vary, but all human settlements throughout the world — from the tiny hill towns of Italy to the sprawling metropolii of London or Tokyo — have some **analogue** of such a system; an analogue which modulates with climate, income levels, cultural patterns, etc. of the society concerned

There are two important facts about the workings of these systems. The first is that each element consists of both covered spaces and open-to-sky spaces. This fact is of fundamental significance because most developing countries are in tropical climates where essential activities can and do take place outdoors. Provided of course, that privacy is reasonably assured, cooking, sleeping, entertaining friends, children's play, etc. need not be exclusively indoor activities, but can function effectively in an open courtyard. In Bombay, for instance, we estimate that at least

**analogue
of a system**

**Covered space
and Open to sky
spaces**

Elements are
**mutually
inter-dependent**





neighborhood spaces [C]



Enclosed room

Verandahs

Pergola covered terraces

Tree shaded courtyard

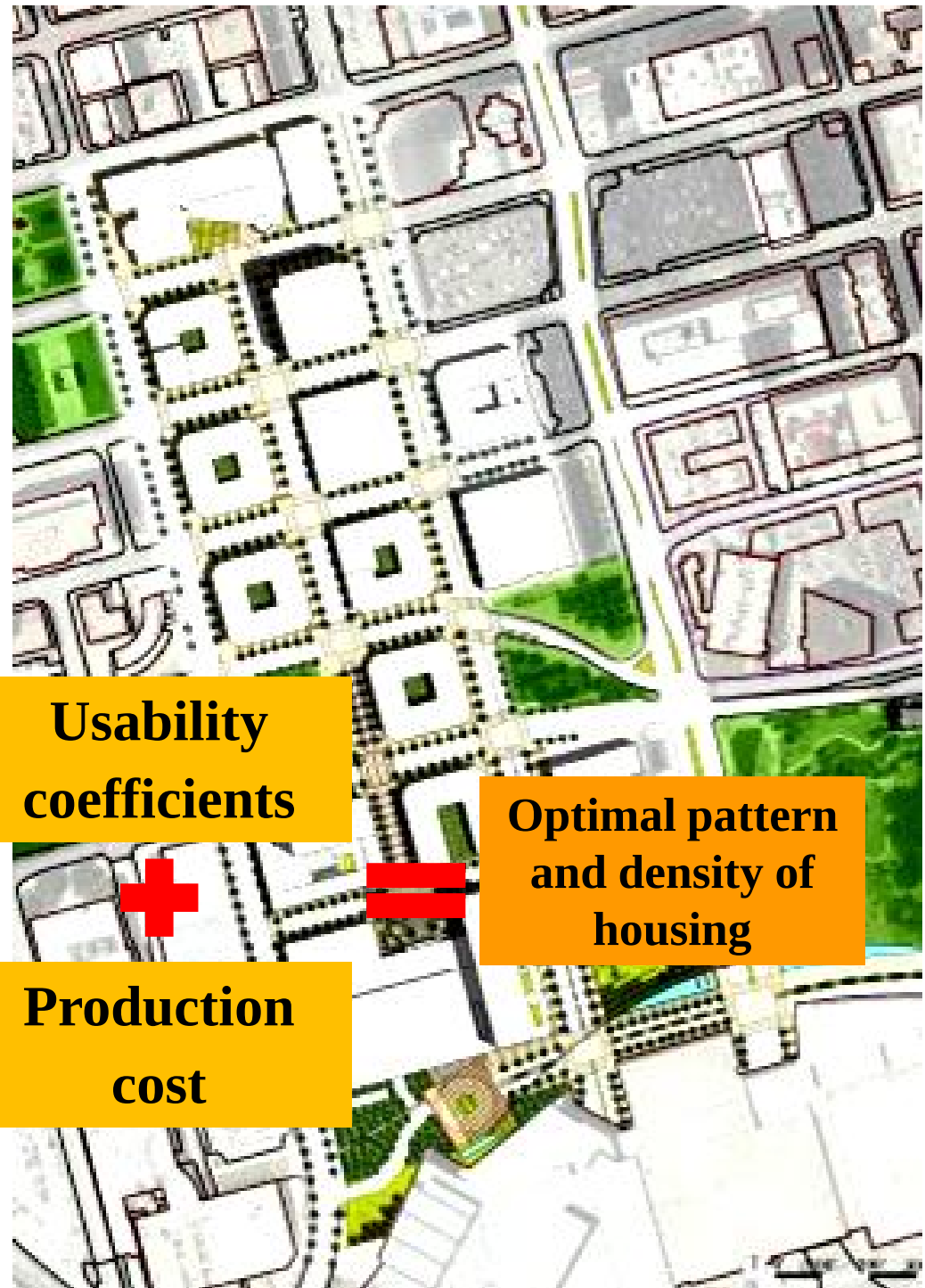
Open to sky space

**Usability
coefficients**



**Production
cost**

**Optimal pattern
and density of
housing**



Use of open to sky spaces

In using open-to-sky spaces, the territorial privacy of families is of decisive importance, for as the surrounding buildings get taller, these spaces become more and more restricted in function. A ground-floor courtyard can be used by a family for many purposes, including sleeping at night. Two storeys, and one can still cook in it. Five storeys and it's only for children to play in; ten storeys, it's a parking lot. The old indicators of so many square metres of open space per 1000 persons are too simplistic and crude to be useful. We must *disaggregate* these numbers, both qualitatively and quantitatively.

**Territorial
privacy**

**Function
restricted**

**Surrounding
buildings get
taller**

Ground floor

Many purposes including sleeping

Two storeys

Cooking

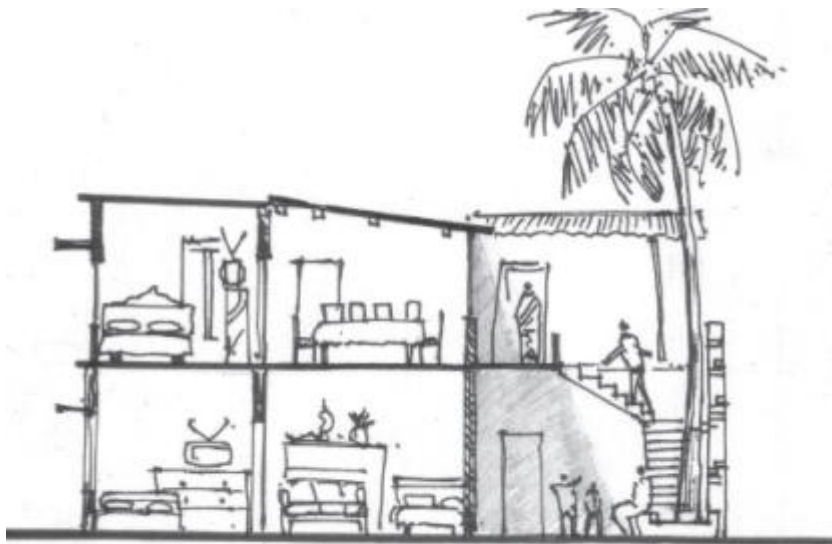
Five storeys

Children to play in

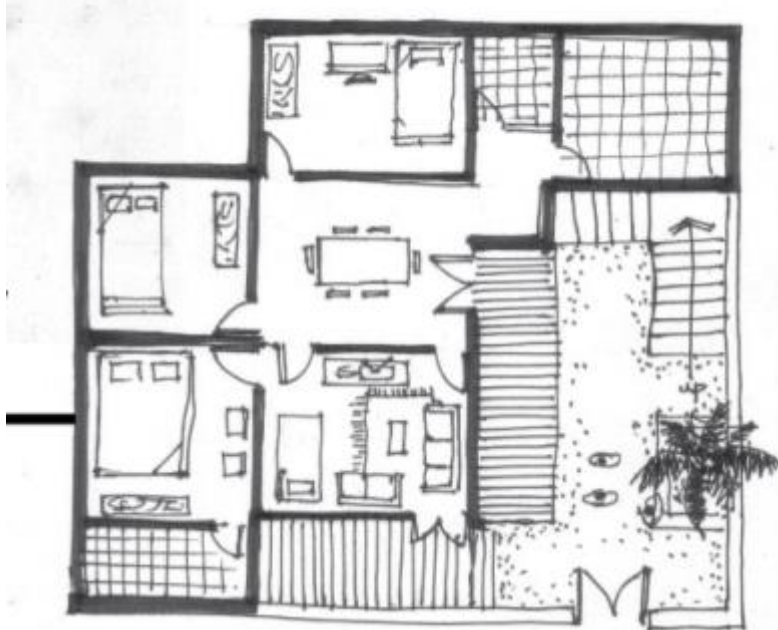
Ten storeys

Parking lot

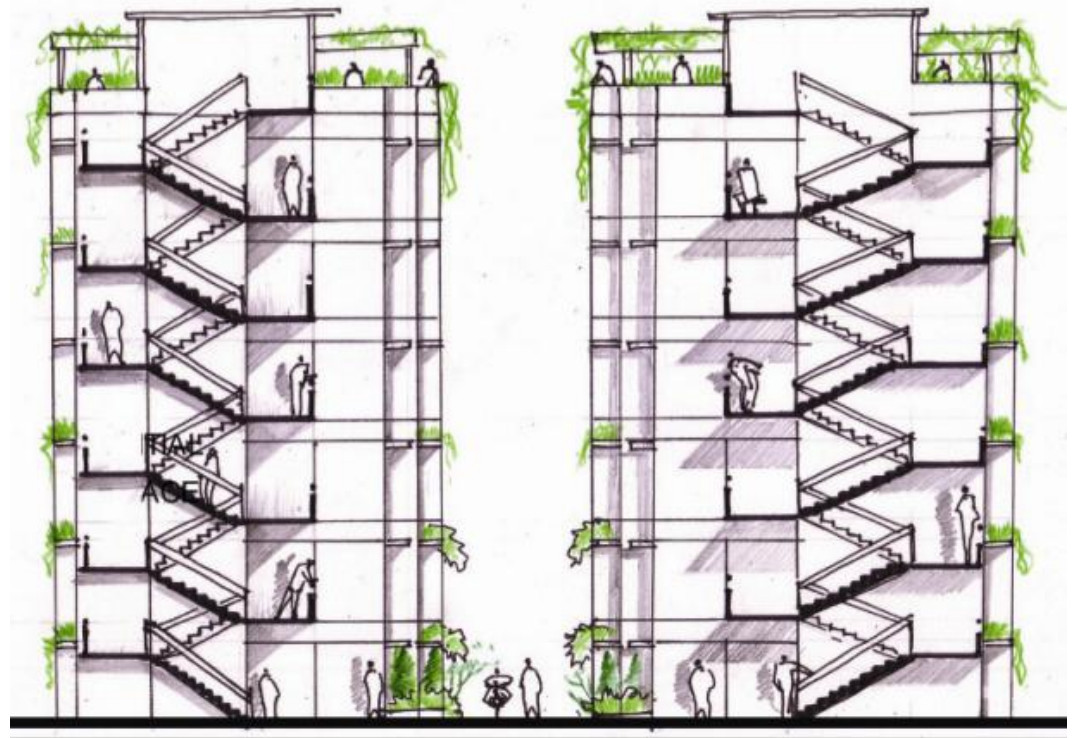
Disaggregate the numbers: breaking up into component parts

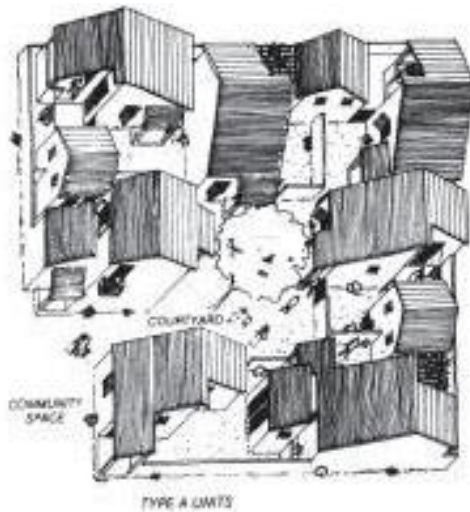


Two stories: Cooking



Five to ten stories:
Children to play in or parking lot



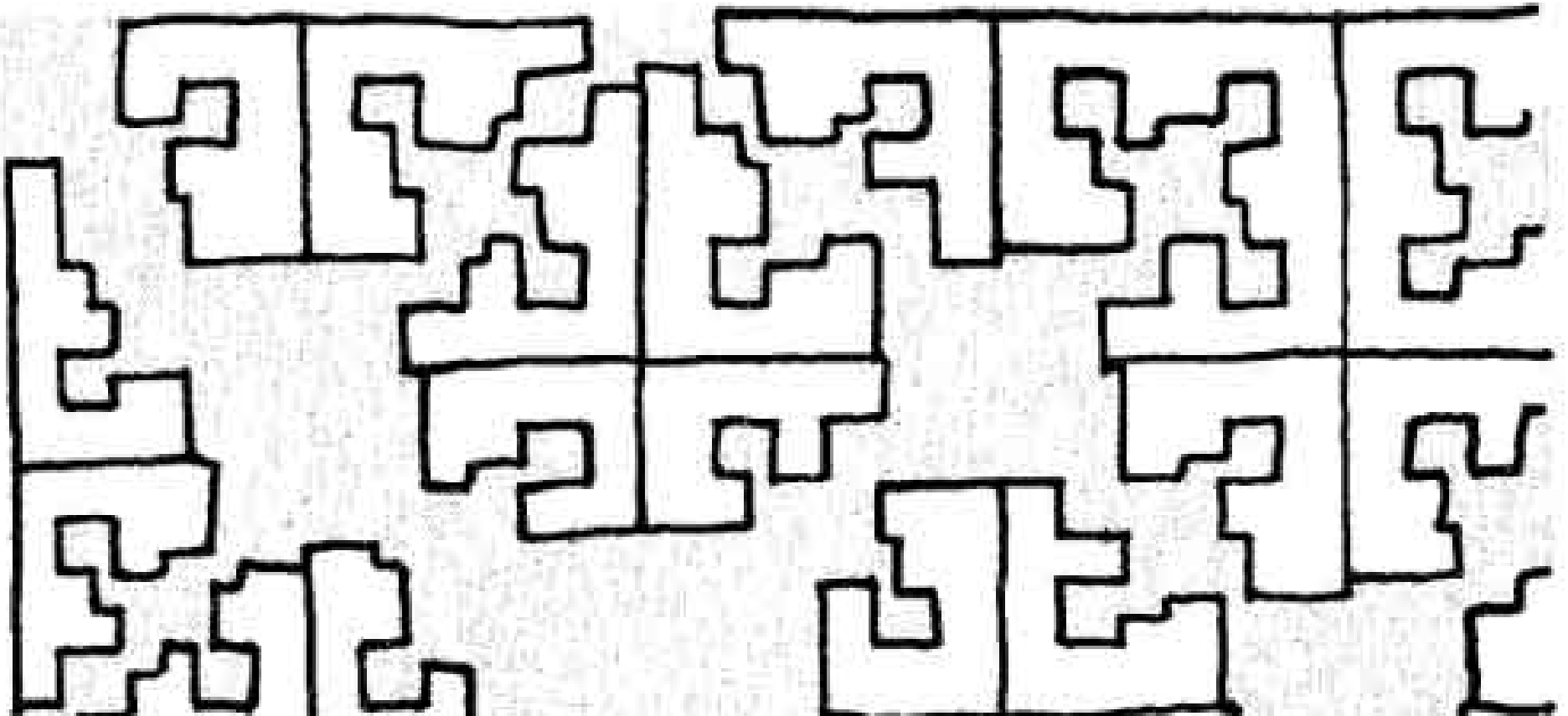


To identify this hierarchical system, and to understand the nature of these trade-offs, is, of course the first essential step towards providing viable housing. Without it, one is in grave danger of formulating the wrong questions. This misunderstanding is the reason why so many attempts at low-cost housing perceive it only as a simplistic issue of trying to pile up as many dwelling units, (as many cells) as possible on a given site, without any concern for the other spaces involved in the hierarchy. The result: environments which are inhuman, uneconomical – and quite unusable. Their planners have ignored the fundamental principle, namely, that in a warm climate – like cement, like steel – *space itself is a resource.*

Today, nearly the entire building industry in our major cities turns out a product that only the middle and upper classes can afford, forcing half of our society out on to the pavements. In their confusion and desperation, architects and engineers search for “miracle” technologies, rather like the medieval alchemist’s fevered hunt for the touchstone which would convert dross into gold. For too long have we struggled for answers when from the beginning the question has been wrongly formulated in the first place. The problem of housing the vast majority of our urban people is not one of miracle building materials or construction technologies, it is primarily a matter of density, of *re-establishing land-use allocations*.

Re-establishing land-use allocations

equity 03

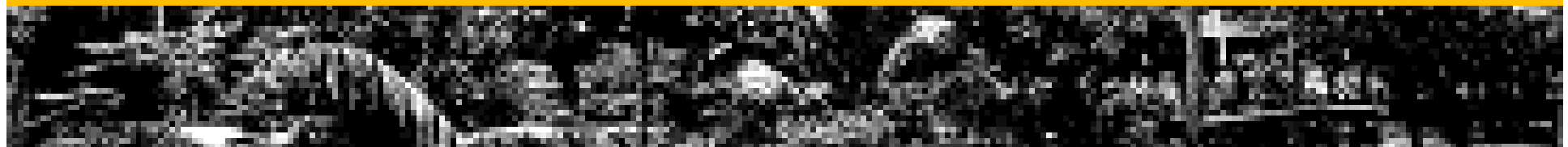






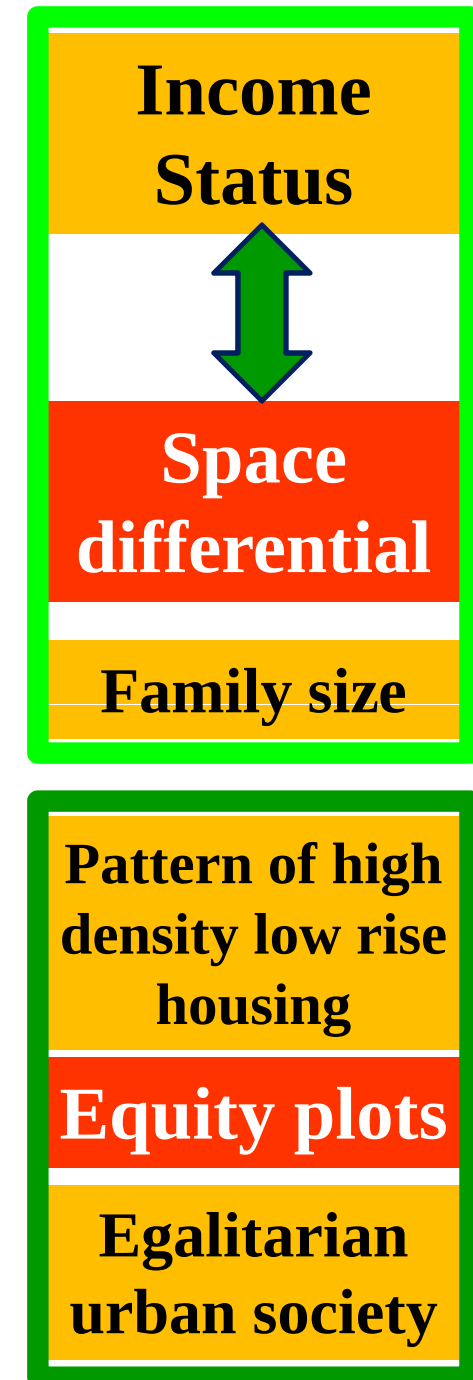


High rises, high expenses, high maintenance



Today the amount of urban space one controls is directly proportional to one's status and/or income; it has no connection with actual family size. Poor people have families at least as large as rich people, often larger. This space differential, therefore, cannot be justified in human terms, only in economic ones. By contrast consider the cities of Australia where almost every family has a quarter-acre lot — no more, no less. Australia is locked into equality — it can never become elitist. The exact opposite is true of most of the Third World. Despite all our rhetoric about social justice and equal opportunity, **we are locked into inequality.** Our cities make sure of that.

This inequality is directly generated by the grotesque skew of the income profile, yet the pattern of high-density low-rise housing gives us a way out of the dilemma. Plot sizes ranging from 50 square metres to 100 square metres can be viable both for the poorest sections of society (furnished perhaps with just a couple of trees, a tied-up goat, and a lean-to roof) as well as for the affluent.



Crucial advantages of low rise housing

01. It is incremental
02. It has great variety
03. This pattern is sensitive
04. It makes for speedier provision of housing
05. It has much shorter construction period
06. No need of high priority construction materials
07. Renewability
08. Maintenance is easier.

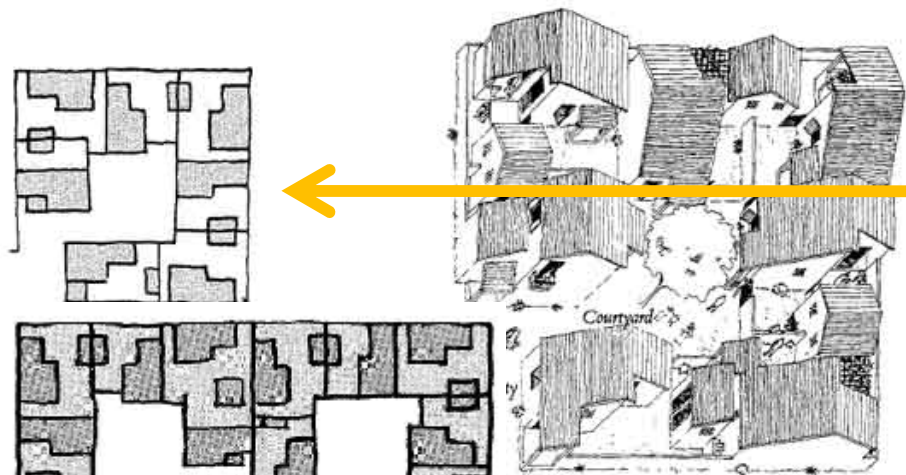
But for the Third World there is one other advantage to this pattern of housing that may prove to be the most decisive of all, and that is *Equity*

A policy of Equity Plots has the added advantage of not pre-determining the social and economic mix either in neighbourhoods or across the city. Most planning today, no matter how liberal the intention, ends up with a rigid caste system of residential areas. Chandigarh is a good example. The reason for this is simple. Since the plots are of vastly different sizes, planners have to decide *beforehand* about their positioning within a sector. In such a situation, a planner cannot place clerks' houses cheek-by-jowl with those of Ministers, and once the plan has been implemented on site, the pattern cannot be changed. So we get cities which are rigid and inflexible. They cannot respond to the social forces constantly at work which make older "un-designed" cities such an organic mix of income groups and communities.

If there ever is a bill of rights for housing in the Third World, it will surely have to include — enshrine! — the cardinal principles:

Incrementality
Pluralism
Participation
Income generation

Equity
Open-to-sky space
Disaggregation



Cluster of
7 houses

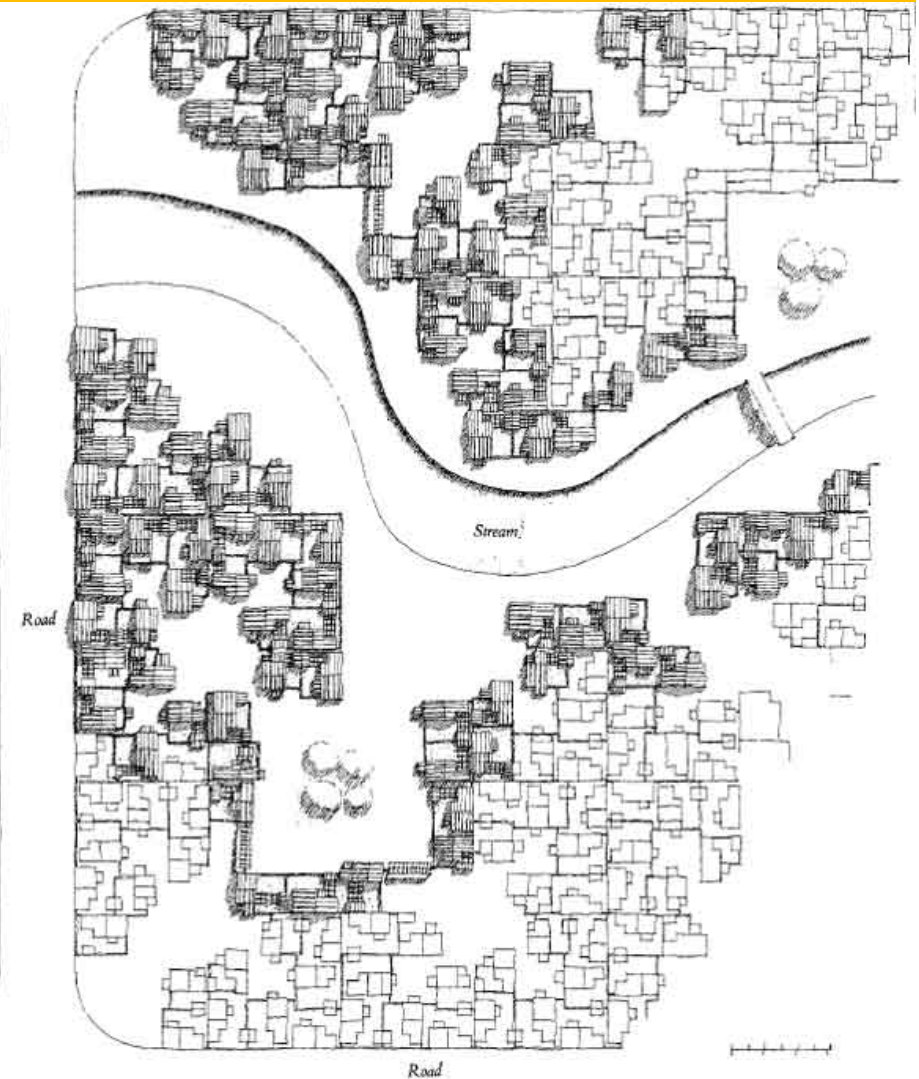
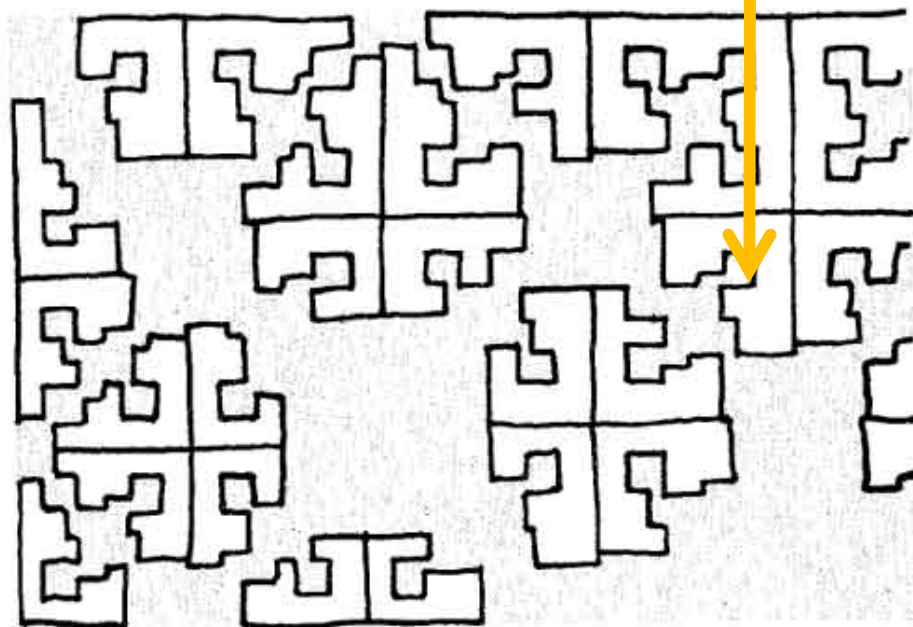
Housing at Belpur, India

Master plan

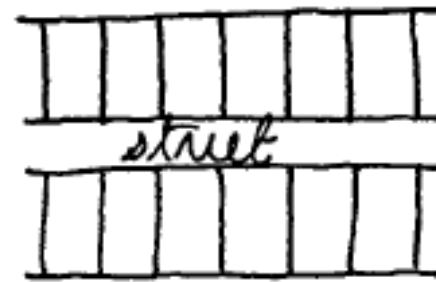


Cluster of
3 x 7 houses

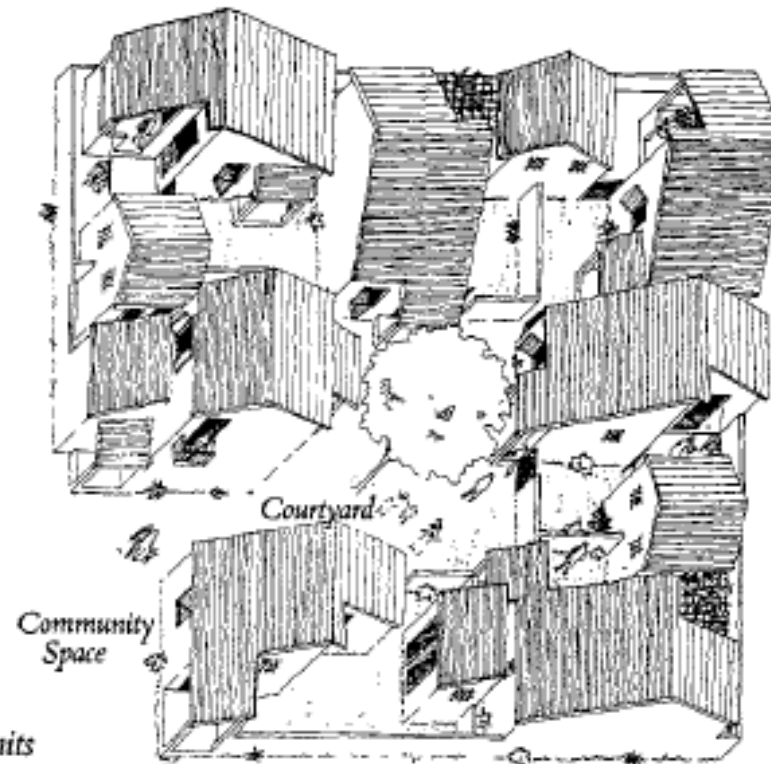
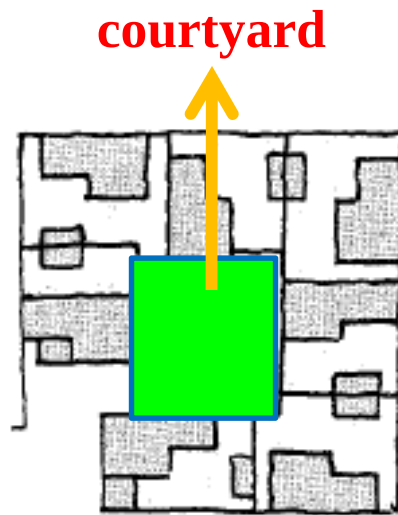
Clusters of
3 x 3 x 7 houses



Usually, low-rise high-density housing takes the form of row-houses, organised along monotonous linear corridors.

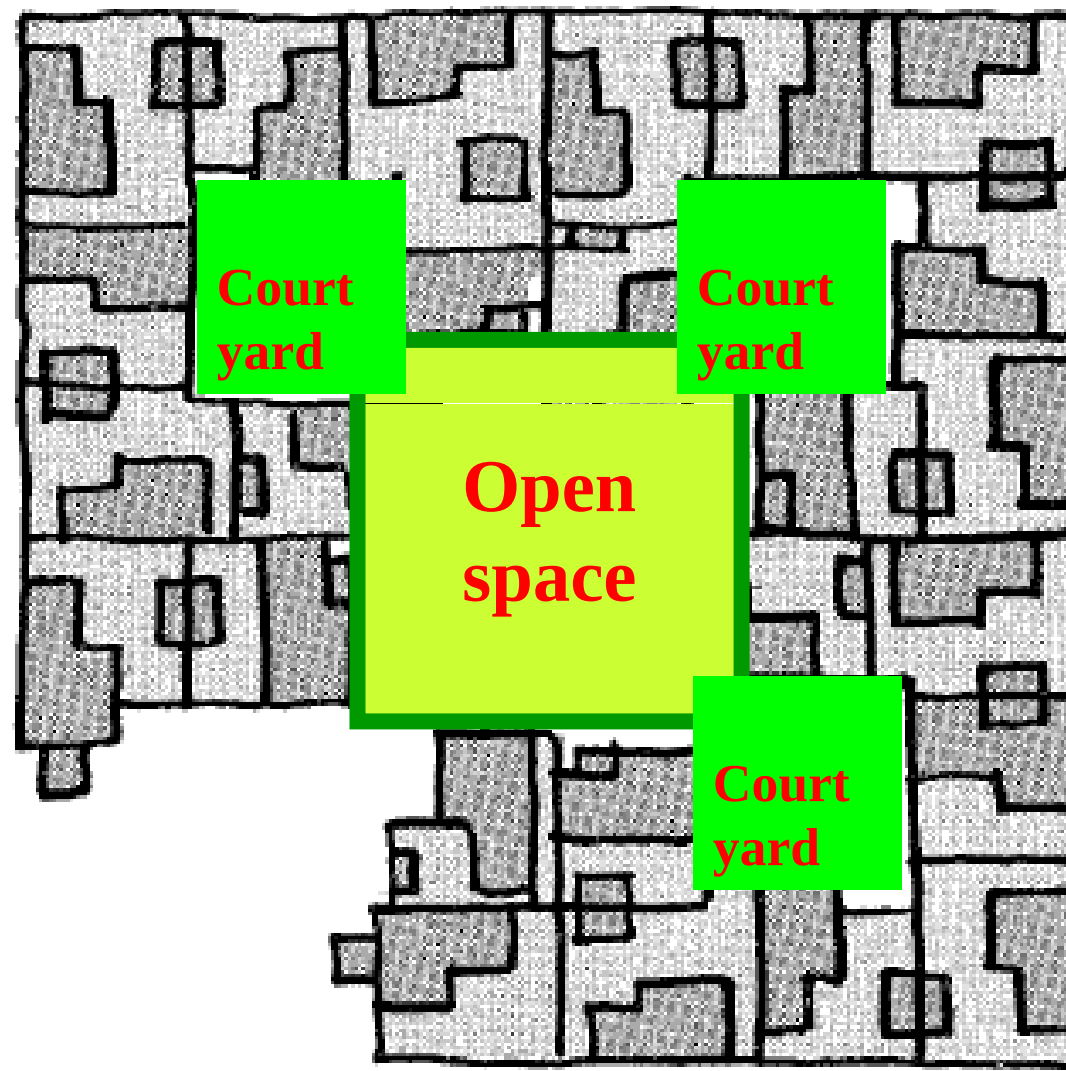


In this case, the units are clustered instead around small community spaces. At the smallest scale, 7 units are grouped around an intimate courtyard 8 metres $\times$ 8 metres.

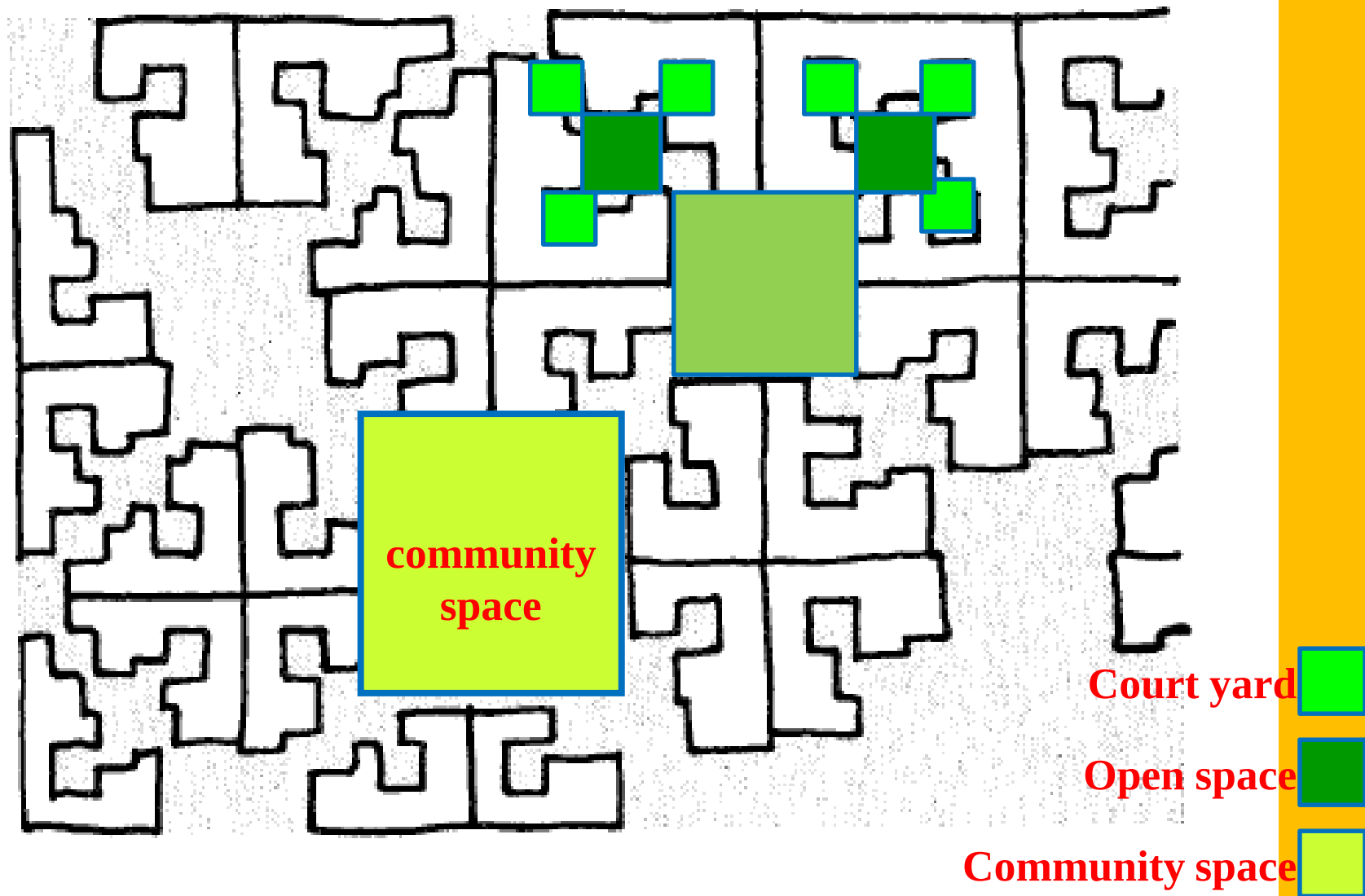


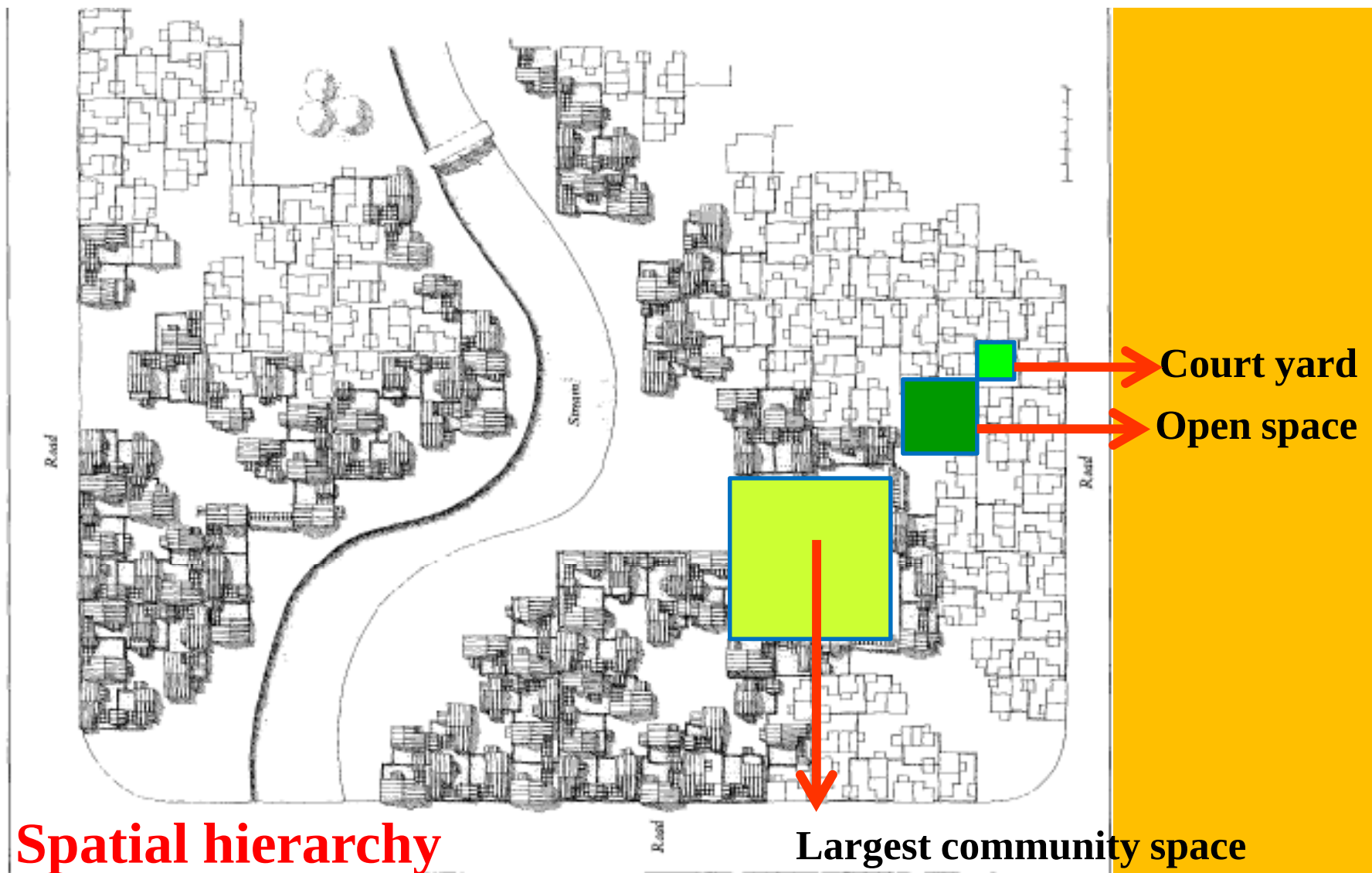
Type A Units

Three of these clusters combine to form a bigger module of 21 houses, surrounding an open space 12 metres $\times$ 12 metres.

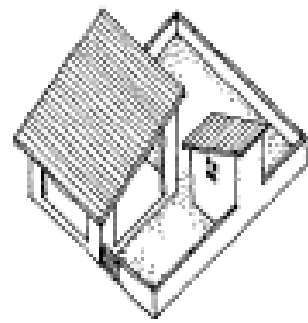


Three such modules interlock to define the next scale of community space — approximately 20 metres $\times$ 20 metres.

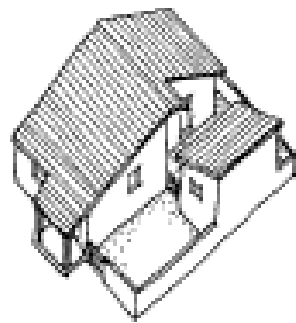
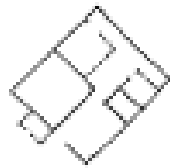




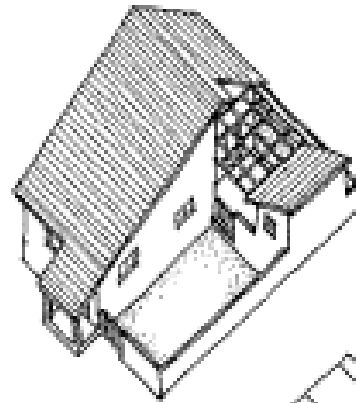
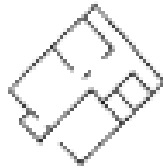
This spatial hierarchy continues until one reaches the largest neighbourhood spaces where primary schools and other similar facilities are located. Down the centre runs a small stream which drains off surface water during the monsoons.



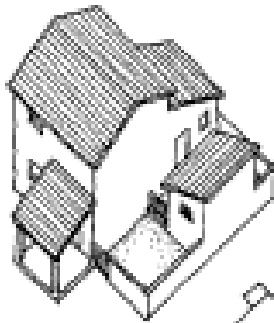
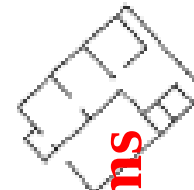
House Type A



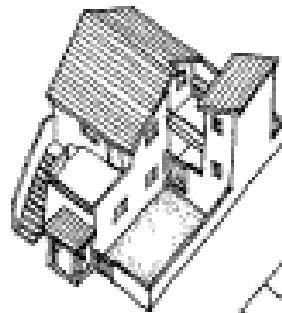
House Type B



House Type C



House Type D



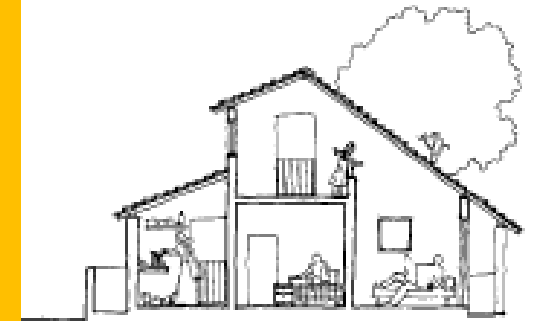
House Type E



Typology of house forms

Incrementality

**Income
generation**



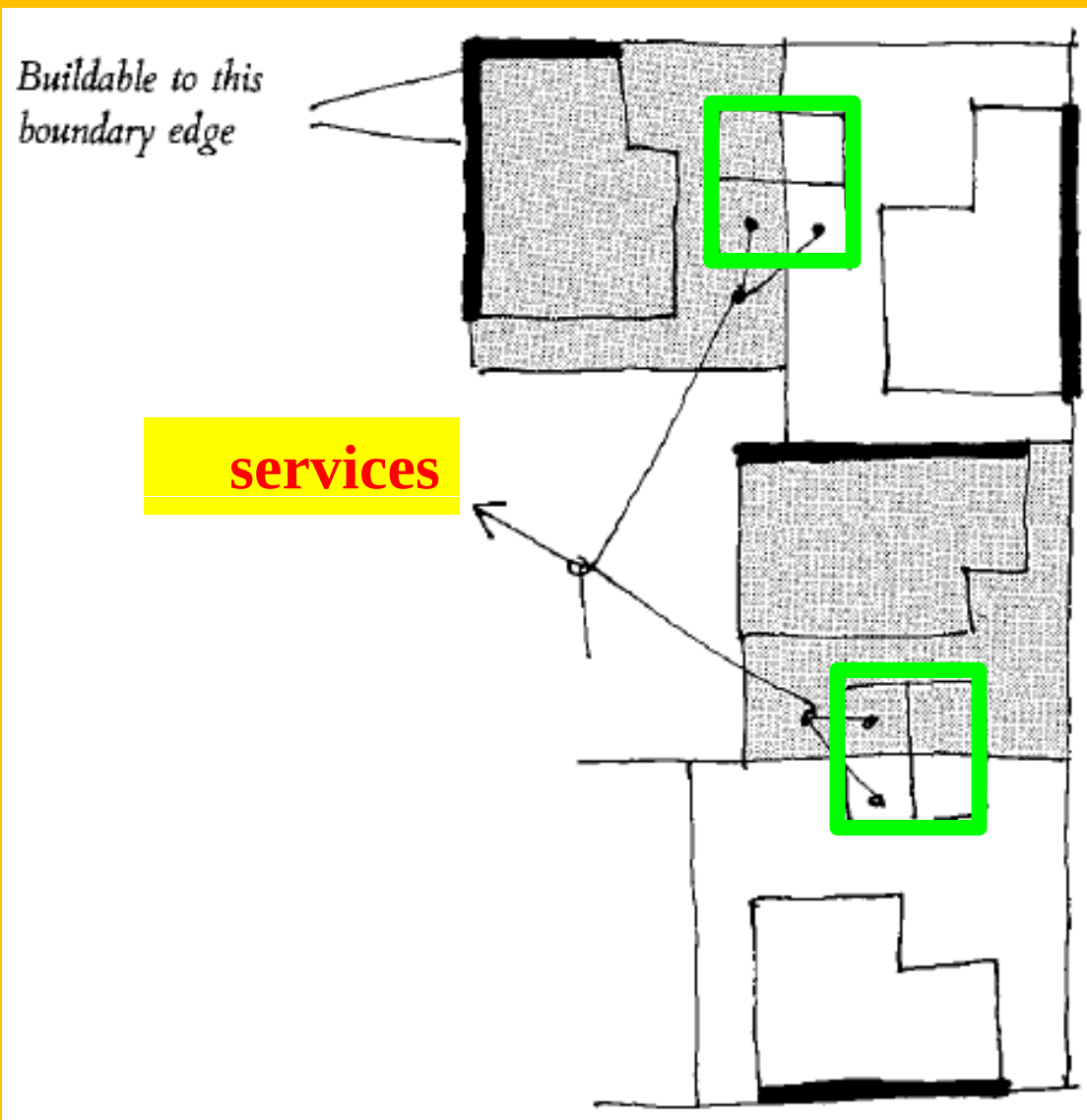
Section: Type B



Elevation: Type C



The typology of the houses forms two sets. Within each set, the houses can grow incrementally to the next state of development, as family income increases.



Hierarchy of space

In Belapur, the house sites are arranged with the toilets located in pairs to save on plumbing and sanitation costs. For each site, the main structure of the house can abut the boundary on two clearly specified edges — in a pattern which ensures that it will be free-standing with respect to its neighbours. No windows are allowed in these walls, so as to protect the privacy of all concerned.



type C

type B



only pedestrian access
 vehicular access

0 5 10 20 m



Detail of typical module



Basic module
45m by 45m



Four modules



Sixteen modules



0 20 40 100 m



green space

house



road

entry court

house

private garden

community green

private garden

house A1

entry court

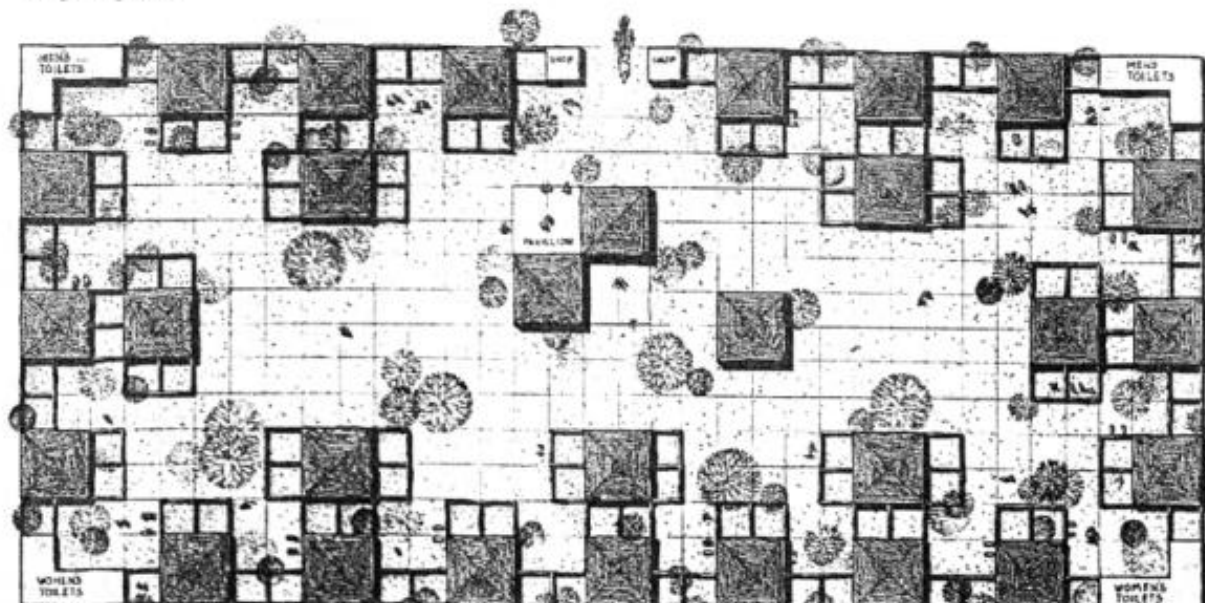
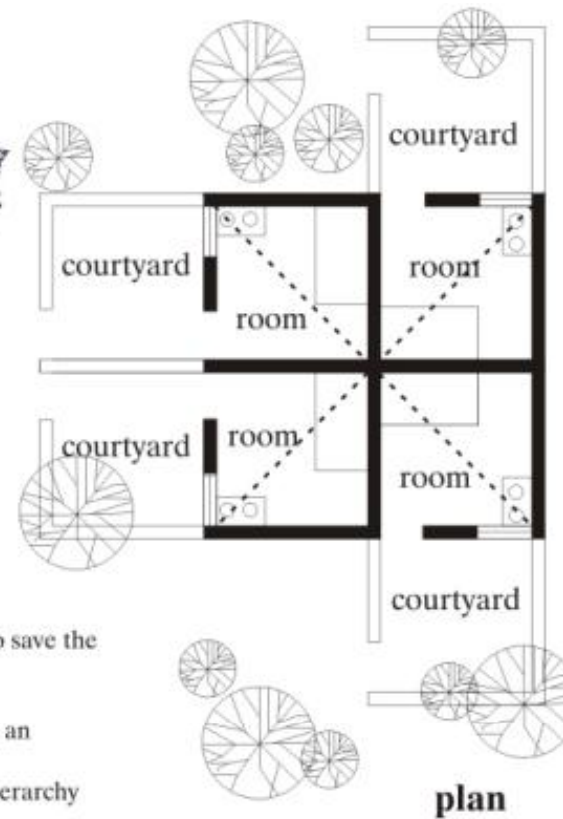
road

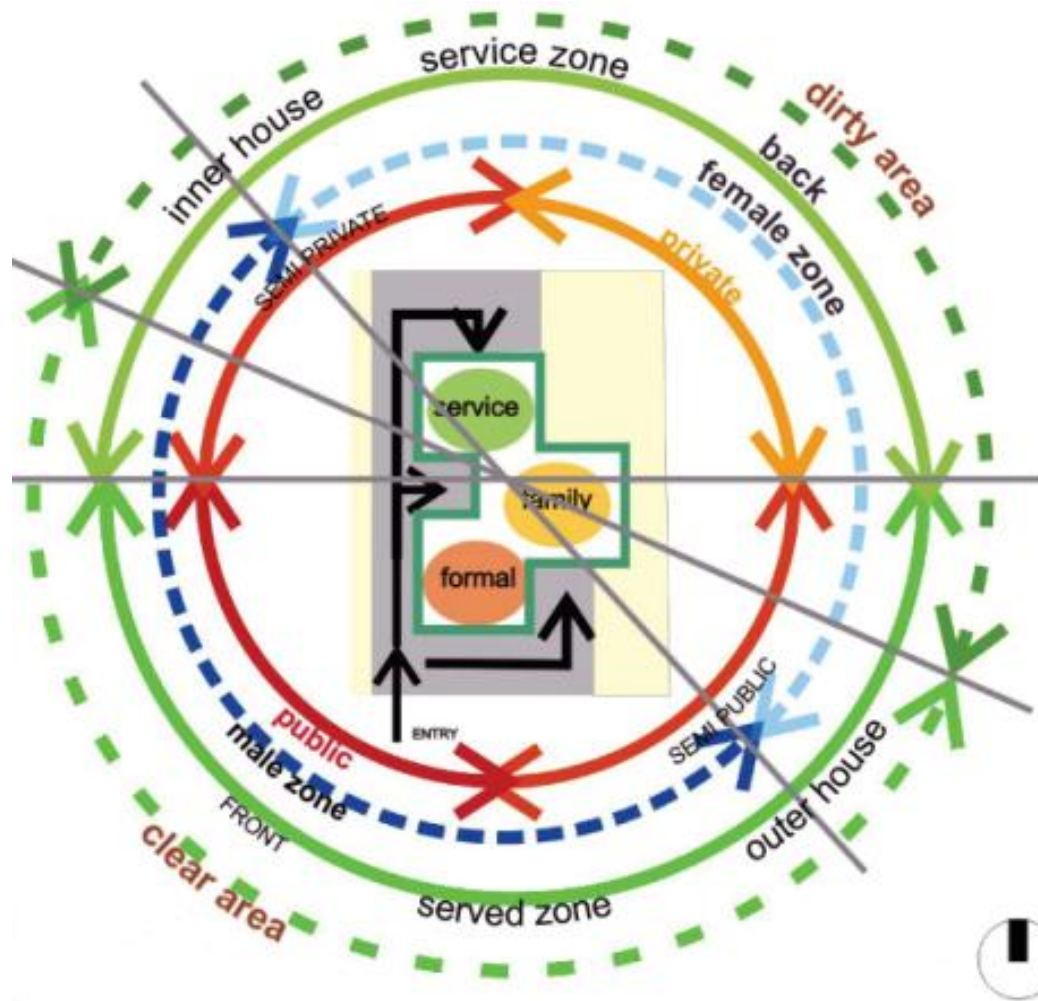
TITAN TOWNSHIP



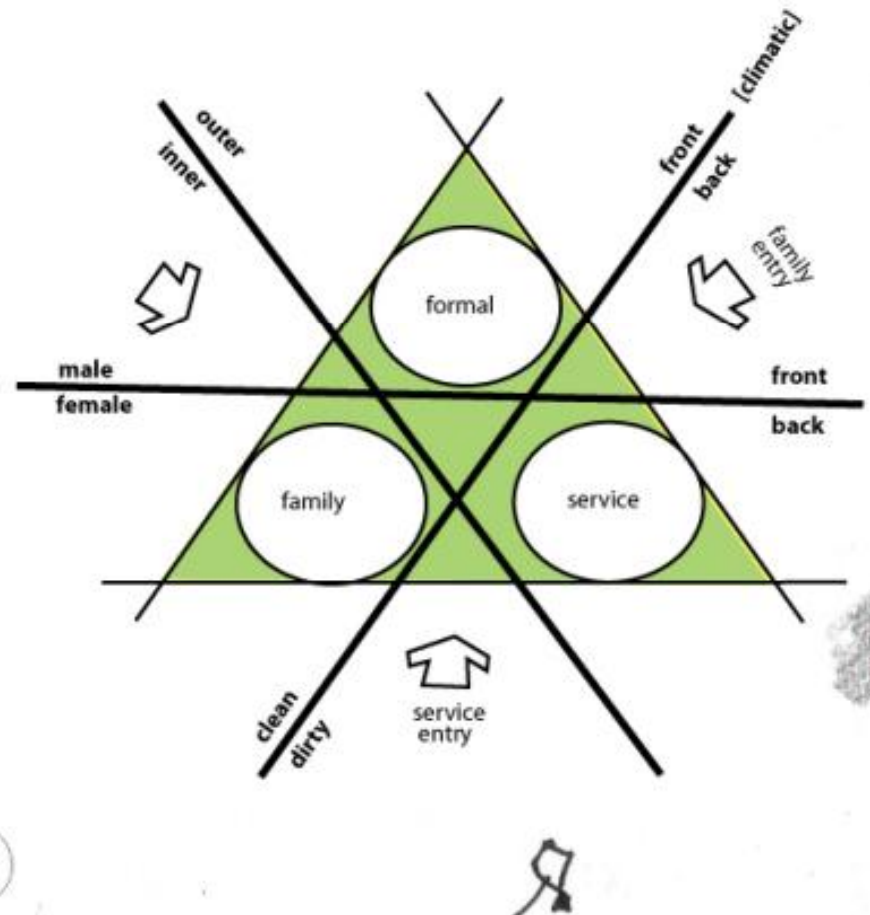
squatter housing: 1973 Charles Correa

- four units are placed together under a paramedical roof to save the construction cost.
- each room is entered through a court yard.
- this court is a open space, which, in a warm climate, act as an additional room.
- the units are arranged in a pattern of cluster to create a hierarchy of open spaces.

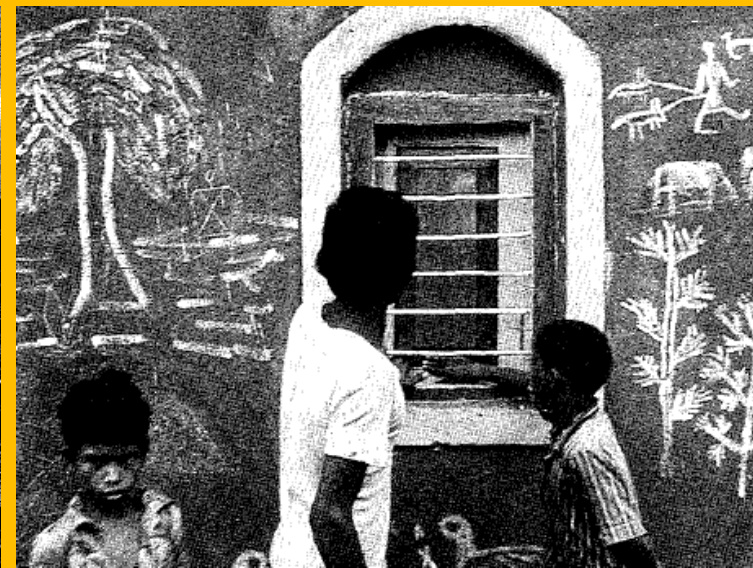
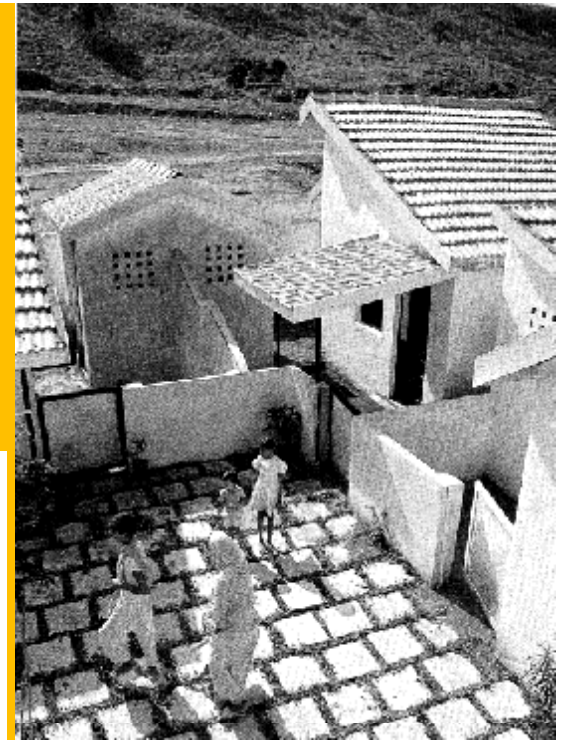




ORGANIZATIONAL MODEL OF THE HOUSE

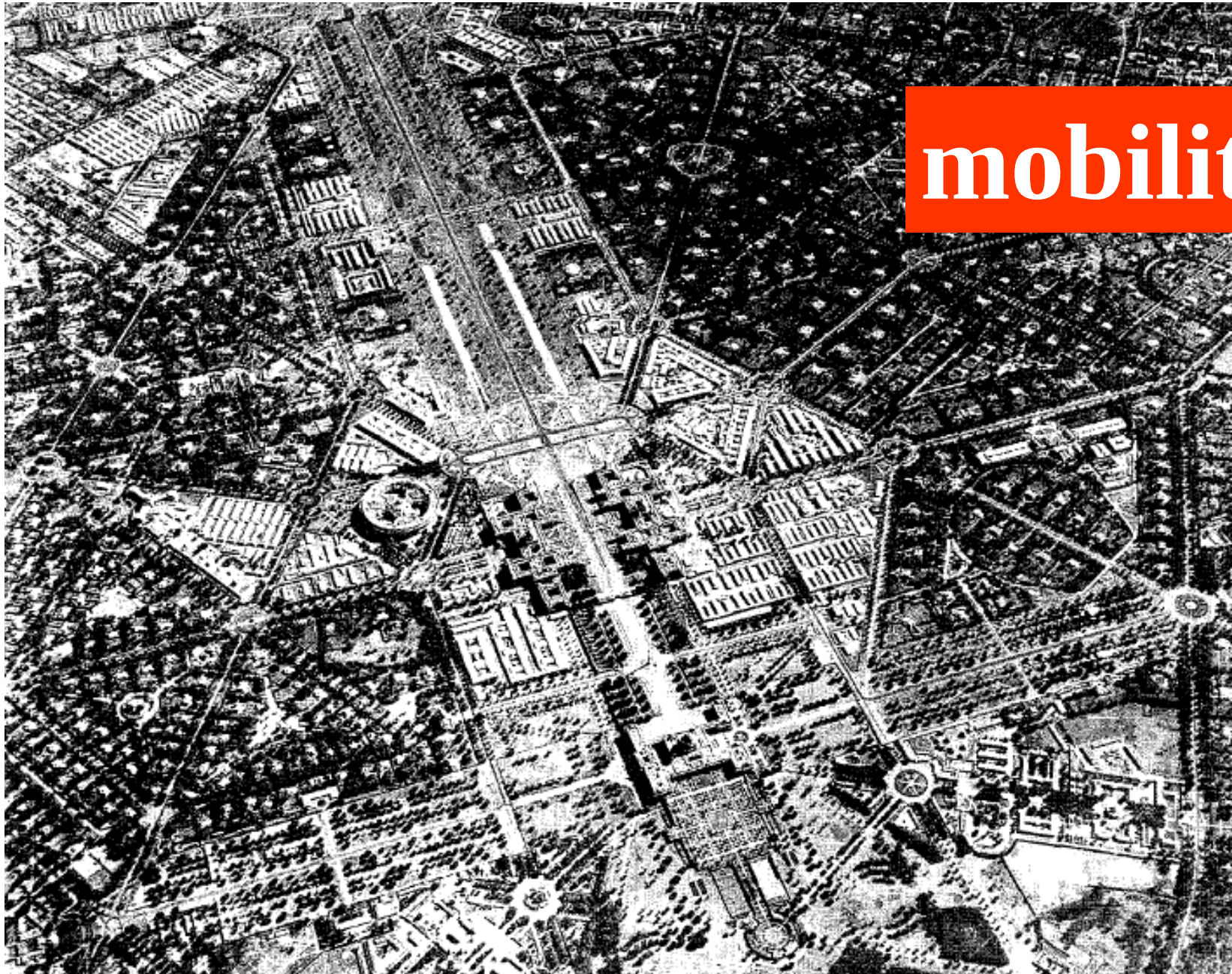


These house plans and drawings are merely indicative. Construction of the units is simple enough to be undertaken by local masons and *mistris*, with the active participation of the people themselves. In time, the occupants will add their own overlays of colour and symbols, colonising it with their lifestyles.



04

mobility



Probably one of the most vivid images of urban China is of a wide avenue of trees, filled with people in blue clothes all cycling to work. It is a happy, positive image, one which underlines the primary importance of the bicycle in developing countries. For a low capital investment, the bicycle provides an individual mobility with no concomitant costs for petrol or other fuel. Bicycles are energy-sane and pollution-free

The landscape of our Third World city should certainly include special bicycle paths, so that they are protected from automobiles and in turn do not interfere with the pedestrians. Where already available, these paths have proven very popular; but even in cities where they do not exist, people cycle great distances to work — so fundamental is the bicycle as a mode of transport in the Third World.

Yet obviously, as the city gets larger, this viability decreases, both because of the distances involved, and because of the increased density of cross-city traffic clogging the main arteries. Some sort of public transport then becomes advantageous.

Now a public transport system is, almost by definition, a linear function. It is viable only in the context of a land-use plan which develops corridors of high density demand. Bombay, for

Bi-cycle

Individual mobility

**Low capital
investment**

**Energy-sane &
pollution free**

Bi-cycle Path

**Public
transport
system**

Bi-cycle is Fundamental mode of transportation

By locating employment centres *within* housing areas, travel distances and costs can be further reduced and sometimes even eliminated altogether. In this respect, the age-old patterns of work-dwelling mix found all over the Third World (as in the shop-houses of South-East Asia) are far more humane and economical than the exclusive zonal systems introduced by modern town planning.

Different transport systems (from bicycle to bus to train) operate under different cost/capacity constraints. This suggests a hierarchy of transport systems implemented in succession as the city grows. Thus, starting with bicycles and buses in mixed traffic, we can gradually build to more heavy-duty systems, without transgressing at any stage the cost/capacity constraints we face.

There is another important advantage in this approach a single mass transport system generates what is essentially a linear corridor, which usually imposes a severe constraint on site planning. By

**Employment
centers**

Housing areas

**Travel distance
& cost**

**Pattern of work
dwellings mix**

**Different
transport system**



**Different
cost/capacity**

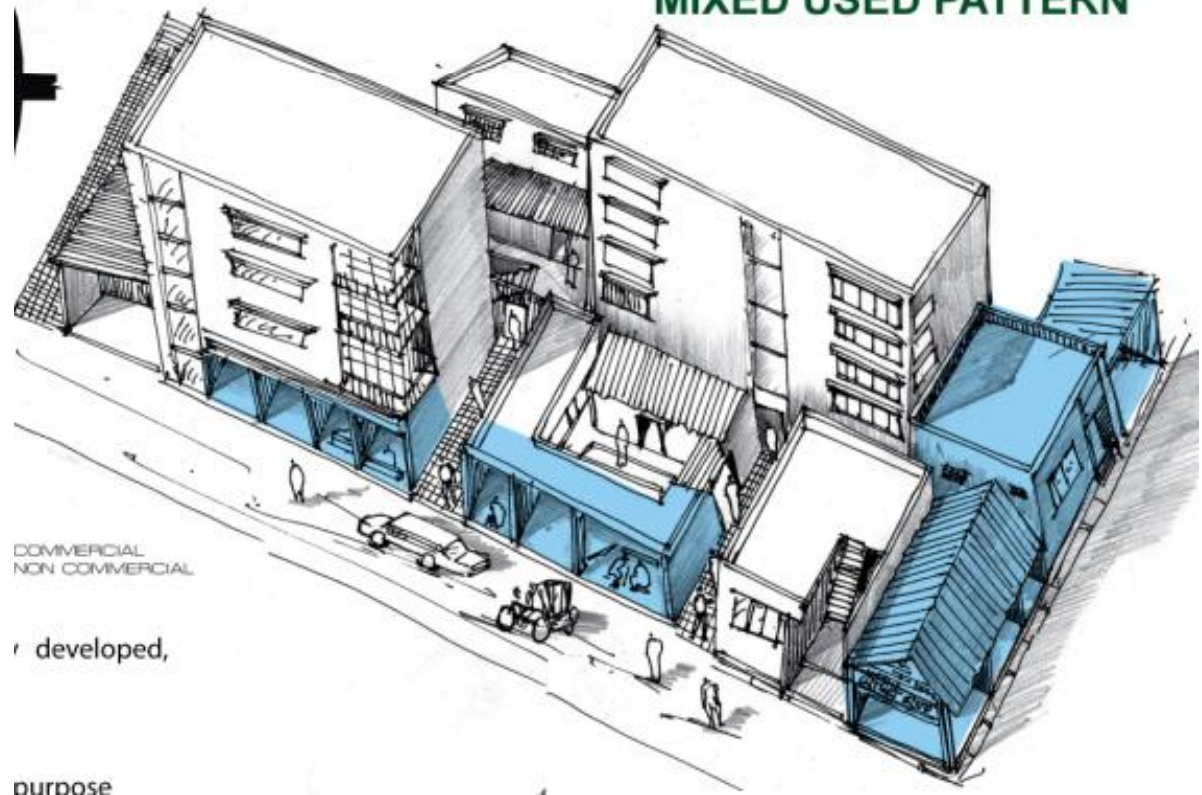


**Hierarchy of
transport system**



Pattern of work dwellings mix

MIXED USED PATTERN



COMMERCIAL
NON COMMERCIAL

developed,

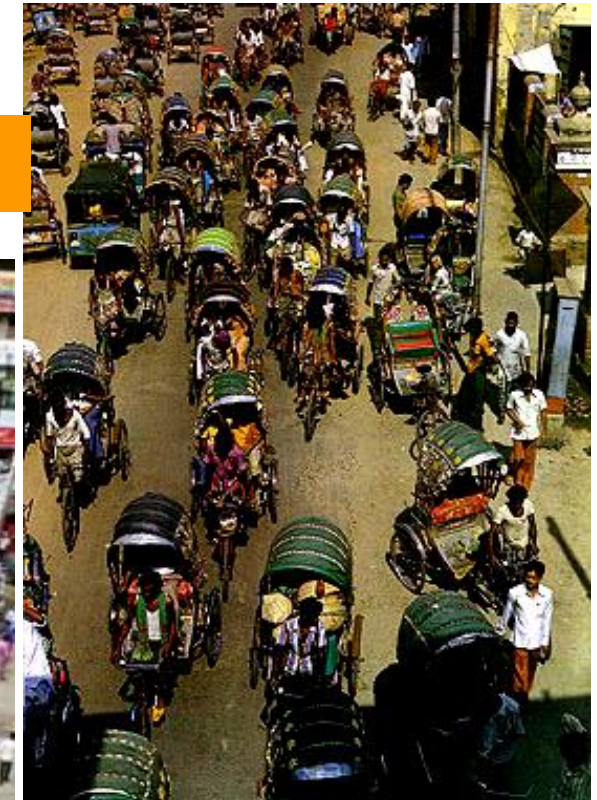
purpose



Shop houses in Bombay: A direct relationship of work and residence which not only avoids commuting but is simple enough to encourage family businesses — which means more economic activity in the bazaar sector, just where the jobs are most needed.



Mode of transportations in & around the city





Mode of transportations

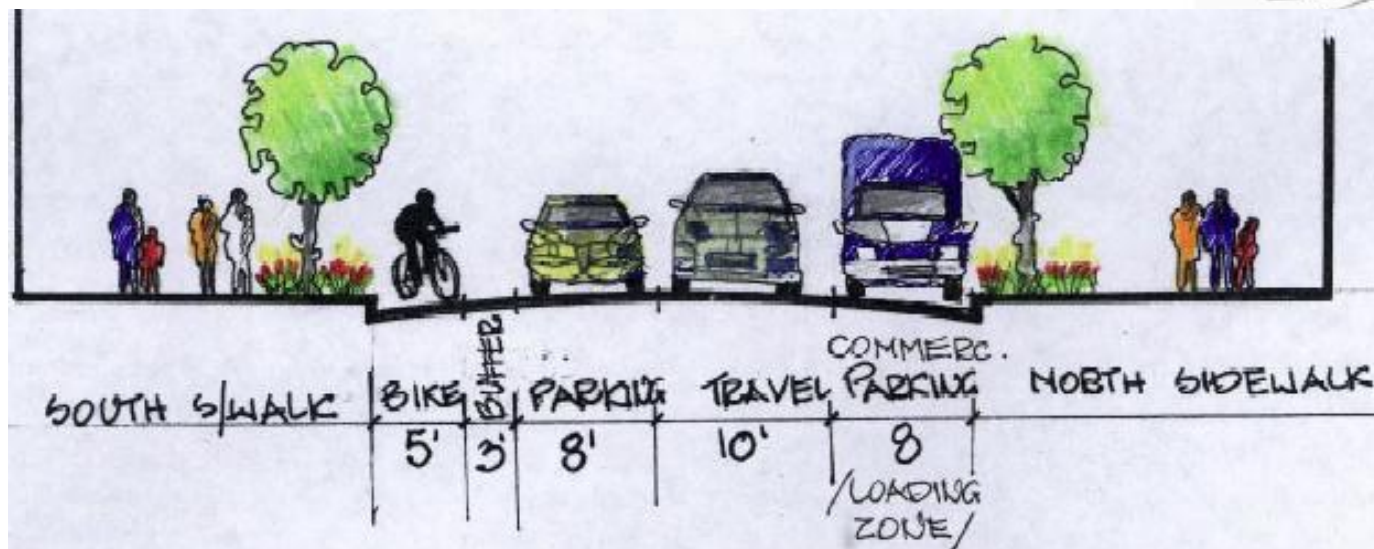


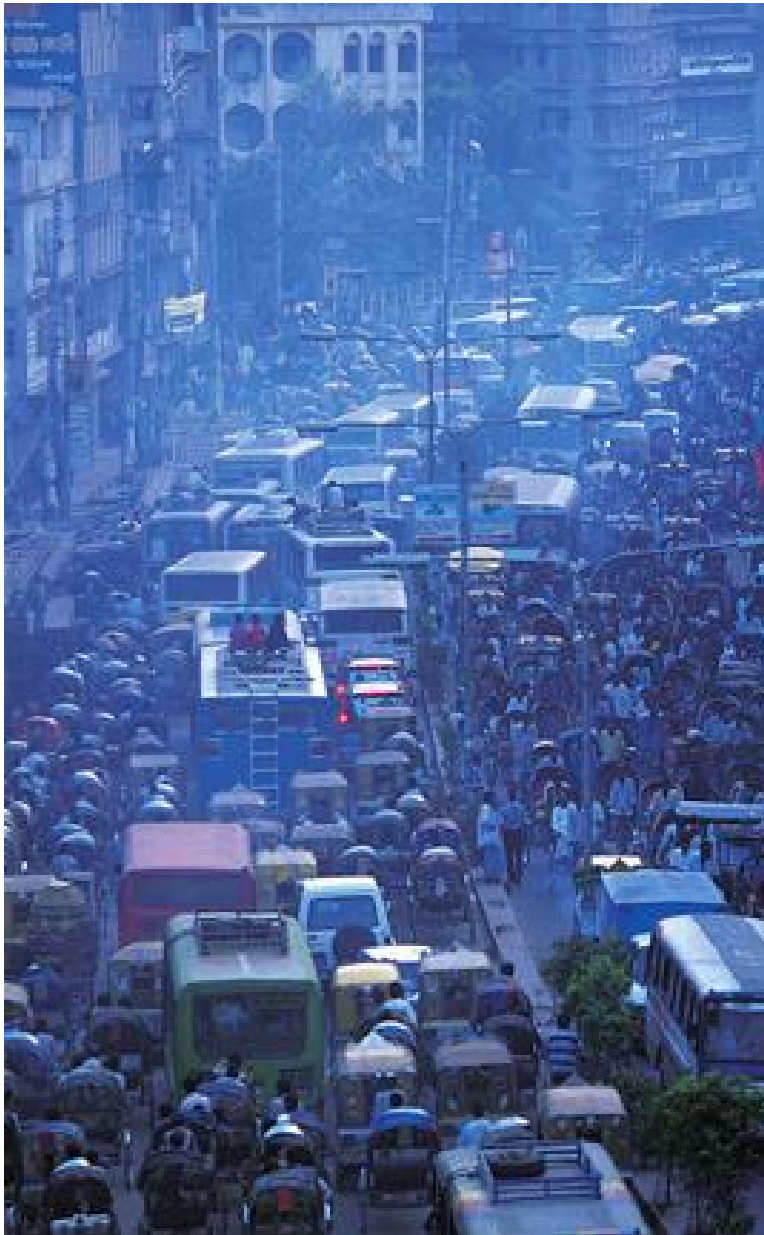


A large, densely populated metropolitan city like Dhaka cannot function without a well-developed system linking all population centers by fast, safe and cheap means of transport..



In 2004, the World Bank funded a \$177 million Strategic Transport Plan to improve transport services for the city's 10 million residents. This program is upgrading the city's extensive transportation network by developing an updated urban transport policy and a long-term strategic plan; encouraging improved inter modal transfers; resurfacing major urban corridors as well as non-motorized traffic lanes; introducing improved parking and zoning regulations; strengthening vehicle registration and driver licensing procedures; and phasing out polluting vehicles.





The recommended plan places special emphasis on the integration of land use and improved zoning. A key element of the plan was reducing accidents and improving traffic flow by separating fast and slower moving traffic. To accomplish this, the plan created separate lanes for Dhaka's traditional rickshaws and developed more than 175 kilometers of pedestrian footways, resulting in a significant reduction in congestion and collisions. Berger also focused on improving bus service and other inter modal connections.

Dhaka Mass Rail Transit (conceptual)

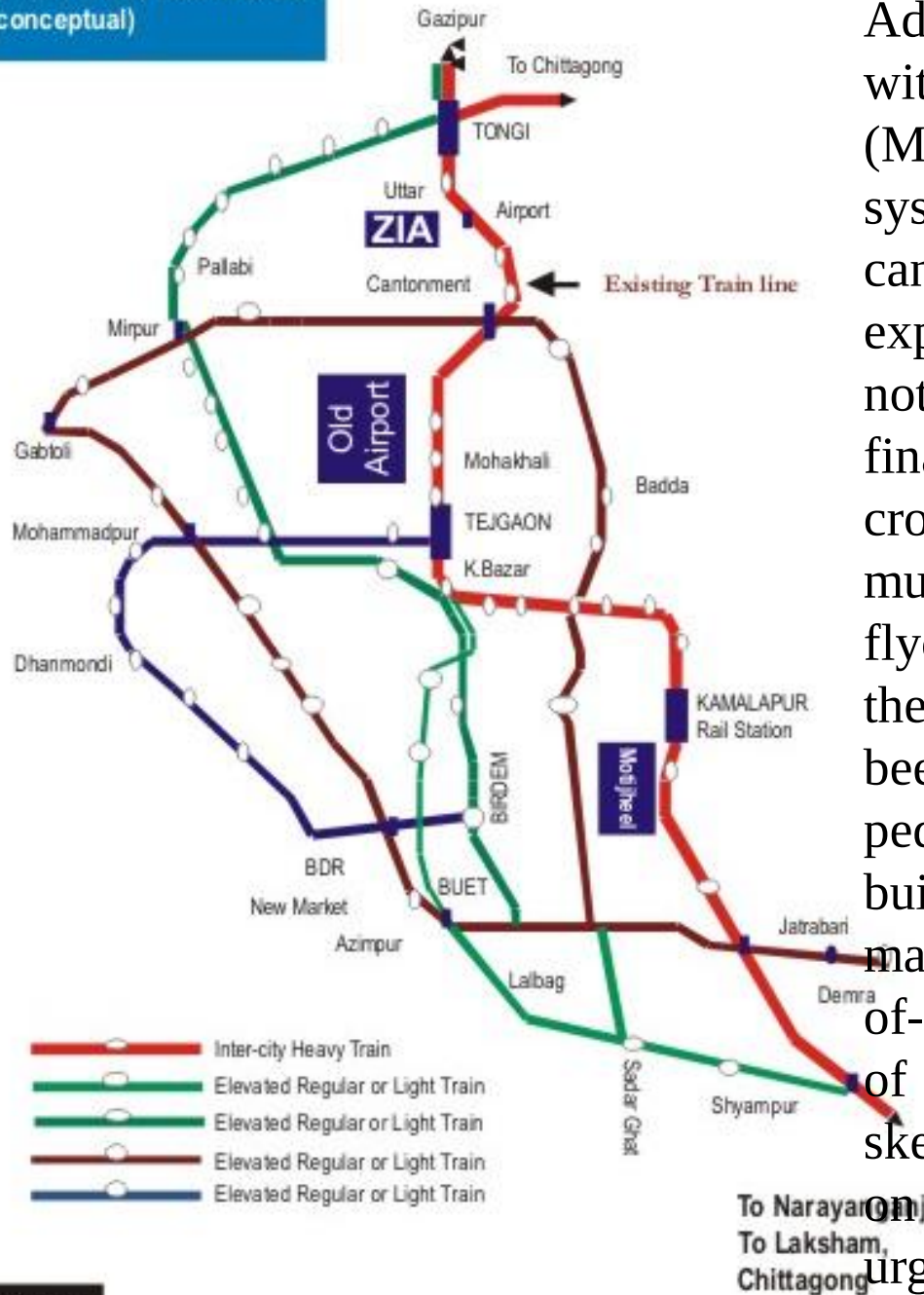


Fig-1

Ad-hoc measures will not suffice; without a well-planned Mass Rail Transit (MRT) system and other public transport systems, traffic jam in Dhaka simply cannot be mitigated. Also, elevated expressway and subway projects must not be taken up before the mass transit is finalized as the expressways normally cross over the elevated train route. We must not repeat the mistake of Mohakhali flyover; had it been 4-6 meters higher, the train line below could easily have been elevated and, in some places, pedestrian over-bridges could have been built to pass below. For the MRT, we may make best use of our existing right-of-way and other infrastructure. Outlines of some probable solutions (absent in the sketchy STP) which should be taken up on a priority basis are given below for urgent government consideration

great CITY....terrible place

Most great cities were terrible places, he said, but their inhabitants were sustained by the myth, the overall image with which a city's identity is inextricably fused.

The key to handling urban expansion, Correa insisted, was not decentralization and dispersal to rural hinterlands, but increasing 'the absorptive capacity' of the metropolitan conglomeration.

In Bombay - which has reached a population of 14 million and, like most Indian cities, is set to double in the next 20 years - he said nothing has been done about restructuring the city (not quite true, in light of the New Bombay initiative for expansion in the 1970s).

Disaggregating the numbers...

of kitchens across the city. We need an analagous methodology for producing housing. It would take a highly inventive management to design it, but it *must* come into being. It is a crucial pre-requisite of the new landscape

As Hassan Fathy, the great Egyptian architect, once said: “Nobody should design more than 12 houses at a time. If you take the greatest surgeon in the world and ask him to operate on 200 people in a day — he’ll kill them all!”

Political will...

“Cities like Mumbai should be governed along the lines of London and New York, which have an elected mayor living in the city. Accordingly, the mayor can then focus on the development and betterment of the city totally and will be responsible to people and not to his political bosses,” Correa said.

“The public are living in a dehumanizing condition. At times, the government officials behave in a docile manner as was reflected in the aftermath of the recent Mumbai blasts, when citizens rushed in to carry the injured and dead to hospital.” he added.

There is a syndrome common to most Third World urban centres. Each seems to consist of two different cities, one is for the poor, another (interlocked with it) for the rich. Very often the latter has a special name. In Bombay, for instance, it is called “The Fort”. In Dakar, Senegal, it is “The Plateau” — an astonishingly accurate name, both in terms of geography, as well as the economic/psychological realities of that city.

Although these two worlds occupy quite separate territories, rich and poor enjoy a symbiotic relationship. This relationship is not always understood by the rich, who often overlook the fact that they need the poor to run the city — not to mention their own households. In most of Bombay’s privileged homes, a multitude of tasks, from the daily washing of the car to the weekly laundry, are regularly performed — for a pittance — by squatters. Ironically, if the squatters in Bombay had to pay for regular accommodation, the cost of their services would rise beyond what their employers could afford! (They are an analogue of the “guest workers” of Europe and the Mexican “wetbacks” in Texas.)

The POOR



The RICH

SYMBIOTIC RELATIONSHIP

quite unsuitable for low-income housing. What is needed is not an acquisition act which nets haphazard little parcels of real estate, but one which delivers large chunks of land at a *scale*, and in *locations* which would allow our cities to undertake the re-structuring they so desperately need. In that sense, the act is radical



help support itself. The concept of self-help housing already exists; there must now come into being the concept of and the programme for — the self-help city.

scanning the options....

Correa was adamant that high-rise was not a solution for urban housing provision. For one thing, it had to be financed by banks which then pocket all the equity. But, also, 'doubling the height of a building doesn't double the density', while the open space around high-rises was wasted. Correa advised that 'we must know how big the numbers are' - in terms of dwellings required - because 'it sends adrenalin through our systems'.

The cluster model Correa developed for courtyard housing in India, with private and shared 'open-to-the-sky space', and a built-in capacity for incremental growth, is well known, but he offered no clear models for, say, London. What he did advise was always to judge optimum density in relation to the other crucial factors of economy, culture and lifestyle.

Balanced ecosystem

Recycling of waste products

Appropriate life styles

Indigenous technology

From Polynesian islands to Mediterranean hill-towns to the jungles of Assam, for thousands of years people have been building incredibly beautiful habitat. In fact, if we look at all the fashionable concerns of environmentalists today — balanced eco-systems, recycling of waste products, appropriate life-styles, indigenous technology — we find that the people of the Third World already have it all. Ironically enough, that's the wonderful thing about the Third World: *there is no shortage of housing.* What there is a shortage of, most definitely, is the *urban context* in which these marvellously inventive solutions are viable.

This then is our prime responsibility: *to help generate that urban context*

URBAN CONTEXT

thanks